Vanuatu National Hospital Laboratory Services Handbook

1st Edition · July 2026

Document ID: VNH Laboratory Handbook.v1

Ministry of Health · Vanuatu National Hospital · Port Vila

PREFACE

This is the Laboratory Services Handbook of the Vanuatu National Hospital (VNH) Department of Laboratory Services.

1.1. DOCUMENT CONTROL

Document ID: VNH Laboratory Handbook.v1.docx

Author	**Mr. SERO Kalkie (MLT, VNH Laboratory)**
Contributors	Dr. TARINAVANUE-GARAE Crystal (Pathologist) Mr. PAKOA George (VNH Allied Manager) Mr. TOM Daniel Robert (Dip. MLS) Mr. SEULE Raymond (Dip. MLS) Mr. KALMOR Jeffery (BMLS) Miss. DICK Leiwia (BMLS) Miss. TOANGWERA Lucienne (BMLS) Mrs. SEWEN Susan (BMLS) Mr. BAKON Samuel (BMLS) Mrs. TOSUL Mary Ann (BMLS) Dr. LOWES Hannah
Verified by	Mr. TOSUL Peter, Principal Laboratory Officer
Authorized by	Dr. TARINAVANUE-GARAE Crystal, Pathologist
Issue date	July 2026
Version date	July 2026
Next review date	July 2027

Do not use this handbook beyond its next review date. The Vanuatu National Hospital Laboratory Department reserves the right to amend, omit, or add to the procedures in this handbook at its discretion.

1.2. HOW TO USE THIS HANDBOOK

This handbook gives pre-analytical and post-analytical information and guidance to laboratory service users when requesting tests available at the VNH Laboratory Department, including:

- Laboratory contact details and opening hours
- Instructions for completing sample and request form information
- Details of services provided, including a current test directory
- Handling and transporting samples to the laboratory

- Point-of-care testing
- Specimen collection procedures and specific sample requirements

It is intended for use by clinicians, nurses, and other VNH staff who request, collect, or handle laboratory specimens.

1.3. HANDBOOK UPDATES

The VNH Laboratory Department intends to update this handbook on a regular basis. Comments, suggestions, and corrections are encouraged and should be directed to the relevant Departmental Officer in Charge, the Principal Laboratory Officer's Office, and/or the VNH Pathologist.

CONTENTS

2. INTRODUCTION

The Laboratory Department is part of the Vanuatu National Hospital (VNH). Its staff aim to provide a quality-assured and accessible service to VNH laboratory users, to promote the well-being of the people of Vanuatu. The VNH Laboratory comprises the following departments:

- Phlebotomy
- Haematology
- Serology
- Biochemistry
- Molecular
- Blood Bank and Transfusion
- Microbiology
- Tuberculosis (TB)
- Histology
- Cytology
- Inventory / Stores
- Reception

2.1. LABORATORY TEAM

Approximately 80% of laboratory staff are formally qualified, holding a Bachelor or Diploma in Medical Laboratory Science. The remainder are on-the-job trained staff with several years of experience.

The Principal Laboratory Officer (PLO) is the most senior position and is responsible for managing the Laboratory Department and overseeing all staff activities. A Quality Officer and Operational Supervisor are responsible for quality control, equipment calibration, and maintenance, and on a routine basis also perform the same duties as other medical laboratory officers. An Officer In-Charge leads each laboratory section and, likewise, performs routine testing duties as well as managing their section. A Laboratory Health and Safety Officer oversees laboratory safety.

A medical Pathologist also attends to the services of the Laboratory Department.

2.2. MISSION STATEMENT

The laboratory section's mission statement was produced in conformity with the Ministry of Health and National Medical Laboratory as to provide a quality assured laboratory service, which supports the delivery of health services at all levels of care prevention, diagnosis, management and surveillance of the Vanuatu's burden of diseases in both urban and rural areas.

Our vision is "To have an integrated and decentralized health system that promotes an effective, efficient and equitable health service for the good health and general wellbeing of all people in Vanuatu."

2.3. OBJECTIVES OF THE LABORATORY SERVICES

- Ensure accuracy of test results.
- Ensure all test results are released to clinicians on time.
- Ensure availability of safe blood and blood products.
- Ensure the requirements and expectations of our customers are satisfied.
- Provide staff with opportunities for professional development and encourage involvement in decision-making.
- Provide ongoing laboratory support to both horizontal and vertical National Health Programs.
- Provide the knowledge, competencies, and skills to support packages of care and maintain professionalism among laboratory personnel.
- Establish appropriate, standardized laboratory designs and infrastructure for each level of care.

- Provide standard packages of laboratory tests at each level of the Vanuatu health system, and promote their rational use for prevention, diagnosis, treatment management, and surveillance.
- Establish and maintain a quality management system to ensure quality laboratory services.
- Promote and strengthen research and development.
- Work towards ISO certification by 2030.

3. LABORATORY DEPARTMENT CONTACT DETAILS

Address: Laboratory Department, Vanuatu National Hospital, PMB 9013, Port Vila

Switchboard: +678 22100

Laboratory Section	Contact	Officer In-Charge
Reception & Phlebotomy	Ext 1954 (VOIP)	Mr. ESCAR Avock (A/Officer in charge)
Haematology	Ext 1949 (VOIP)	Mr. TOM Daniel Robert
Biochemistry	Ext 1949 (VOIP)	Mr. Samuel Bakon
Blood Bank Counselling room	Ext 1968 (VOIP)	Mr. BULEMARU Ray
Serology and Molecular	Ext 1950 (VOIP)	Mr. KALOMOR Jeffery
Microbiology	Ext 1950 (VOIP)	Mrs. TOSUL Mary-Ann
Tuberculosis (TB)	Ext 1949 / 1950 (VOIP)	Mr. SEULE Raymond
Histology	Ext 1949 (VOIP)	Mr. SERO Kalkie (A/Officer in charge)
Cytology	Ext 1949(VOIP)	Miss. TOANGWERA Lucienne
Inventory / Store	Ext 1949 / 1950 (VOIP)	Miss. GAVIGA Justina
Blood Bank & Blood Transfusion	Ext 1950 (VOIP)	Mr. BULEMARU Ray (A/Officer in charge)
Quality Officer	Ext 1949 / 1950 (VOIP)	Mrs. SEWEN Susan
Principal Laboratory Officer (PLO)	Ext 1949 (VOIP)	Mr. TOSUL Peter

3.1. LABORATORY ADVISORY SERVICE

The Laboratory provides professional advisory services to clinicians, nurses, healthcare workers, and other authorized users regarding laboratory test information, specimen requirements, test interpretation, and blood transfusion services.

Laboratory advisory services may be provided through:

- Telephone consultations through the lab.
- Walk-in consultations at the laboratory.
- Official written communication, where applicable through email to the Laboratory Pathologist and PLO.

To ensure the accuracy, consistency, and quality of information provided, all laboratory advisory services conducted by telephone or walk-in consultation shall be provided by the Head of Department (HOD), Laboratory Manager, or an authorized Senior Medical Laboratory Scientist/Technologist.

4. QUALITY SYSTEM

The Laboratory is committed to maintaining an effective Quality Management System (QMS) to ensure the delivery of accurate, reliable, and timely laboratory services. The QMS is based on the laboratory's Quality Policy and supports compliance with recognised international standards, including ISO 15189. The system includes quality documentation, staff competency and professionalism, quality assurance activities, monitoring of result accuracy, internal and external audits, continual review and improvement, and adherence to ethical practice and confidentiality. All staff are responsible for complying with the Quality Management System and the policies and procedures contained within the Quality Manual.

4.1. INTERNAL QUALITY CONTROL

The laboratory participates in an Internal Quality Assurance (IQA) program through the routine use of quality control materials, instrument monitoring, and regular review of quality control results to ensure the accuracy and reliability of laboratory testing.

Department	Quality control schedule
Haematology	Commercial control performed once daily before routine testing begins; drift control performed three times a day.
Biochemistry	Commercial control performed once daily before routine testing begins; drift control performed every working day after midday.
Microbiology	Control performed on each newly prepared batch of culture media; daily and weekly checks on identification reagents (catalase, coagulase, API20E, Gram stain, etc.) upon opening a kit or as needed; antibiotic discs checked three times a year or as needed.
TB Unit	ZN staining technique controlled weekly; GeneXpert controls run with each test.
Serology	Controls run on each new pack of each test kit (HIV, Hepatitis B, Hepatitis C, Syphilis, RPR, HIV-1/2 confirmation).
Transfusion	Controls performed weekly on antisera and platelets.
Molecular Laboratory	Controls performed on each new extraction, RT-PCR kit, positive control material. Controls run with each PCR.

4.2. EXTERNAL QUALITY CONTROL

The laboratory participates in an External Quality Assessment (EQA) program provided by the PPTC EQA scheme. Sections involved are Haematology, Biochemistry, Microbiology, Blood Bank/Blood Transfusion, Serology, Histology/Cytology, TB, and the Molecular Laboratory.

5. OPERATING HOURS

Period	Service Level	Service Hours
Monday – Friday	Full Services	08:00AM – 12:00PM 13:00PM – 17:00PM
Weekday – Out of normal hours	Restricted Service (primarily emergency department/urgent requests)	On-call 17:00PM – 08:00AM
Weekends and Public holidays	Restricted Service (on-call 24 hours, focused on inpatients and the Accident & Emergency Department)	6:00AM – 6:00AM

During restricted (after-hours, weekend, and public holiday) periods, only one laboratory scientist is typically on duty, so service is limited to restricted tests. Another scientist may be called in if workload requires, particularly for Blood Transfusion, Blood Donation Recruitment, Serology, and Microbiology. Clinicians are reminded to request an urgent test out of hours only where the result will directly affect patient management. Phlebotomy collection can be arranged with on-call staff for weekend collection when needed.

6. REQUESTING TESTS

6.1. HOW TO REQUEST A SAMPLE

Please see Annex for copies of request forms;

- General laboratory request form – Annexure 2
- Crossmatch laboratory request form – Annexure 3
- TB laboratory request form – Annexure 4
- Surveillance laboratory request form – Annexure 5
- Pathology request form – Annexure 8

ROUTINE AND URGENT REQUESTS

Routine and urgent requests are handled differently – please ensure the request form reflects the true urgency. Routine requests MUST be requested during normal service hours. Any routine sample arriving to the laboratory without prior contact with the on-call staff will be treated as non-urgent and stored for analysis the following working day. Urgent requests MUST only be made if the results are needed prior to the next working day and/or WILL directly affect patient management. Urgent requests are handled as such, and abnormal results will be telephoned to the requesting clinician or ward, or handed directly to a second party, depending on the situation.

A) COMPLETE THE REQUEST FORM FULLY

Include the patient's first name, surname, date of birth or age, hospital number (where applicable), gender, specimen type, collection date, date of symptom onset, and the requesting clinician's name and signature. Please complete separate request forms for different sample types.

B) FLAG URGENT REQUESTS CLEARLY

Mark the form URGENT if the result is needed before the next routine reporting cycle and will directly affect patient management. Unmarked specimens are processed as routine.

C) USE THE CORRECT CONTAINER FOR THE TEST

Check the Quick Reference table (Section 8.3) and the relevant department's test table below — the wrong container is one of the most common reasons a specimen is rejected.

D) SEND THE SPECIMEN PROMPTLY

Deliver to laboratory reception during normal working hours, or to the on-call laboratory scientist out of hours. A phone call ahead of an urgent or unusual specimen helps the laboratory prepare.

VERBAL (E.G. TELEPHONE) ADD-ON REQUESTS

Additional tests requested verbally must be made within 3 hours of specimen collection and must include: the additional test requested; the name of the requesting clinician; and the date and time of the request.

OUT-OF-HOURS REQUESTS

Only request urgent analysis out of normal working hours if the result is needed before the next full working day and is likely to directly affect patient management. Specimens without a clear urgency note may be delayed or treated as routine — when in doubt, call the on-call laboratory scientist. Routine tests should not be requested after hours; samples arriving without prior contact with on-call staff will be treated as non-urgent and stored for analysis on the next working day.

6.2. AUTHORIZED ENTRY AND SIGNATORIES

Non-laboratory staff are not permitted to enter the laboratory to collect results or for any other reason without permission from the Officer In-Charge — this protects patient confidentiality and prevents introduction of contaminants.

All laboratory request forms must carry a clearly printed name and signature of an authorized, registered, and certified clinician. Forms without this will not be processed and will be returned to the requesting clinician.

7. SPECIMEN CRITERIA

7.1. MINIMUM ACCEPTANCE CRITERIA

- All requisition forms must be completed and specimens clearly labelled.
- Specimen and request form must both carry a minimum of four patient identifiers:
  - First name
  - Surname
  - Date of birth/age
  - Hospital number (where applicable)
  - Gender
- Specimen type and collection date is recorded.
- Date of onset of symptoms is recorded.
- Requesting clinician's name/initials and signature are present.
- Specimen is submitted in the correct, authorized container for the test.
- Sputum specimens must be received in a biohazard transport bag.

7.2. COMMON REJECTION REASONS

- Requisition form and/or specimen missing the minimum identifiers above.
- Haemolysed sample submitted for biochemistry, FBC, cross-match, or blood culture.
- Mislabelled or inappropriate specimen container.
- Clotted sample submitted for full blood count (FBC).
- Sample drawn from an IV line for FBC or biochemistry.
- Specimen containers for neonatal bilirubin sample not wrapped in aluminium foil.
- Saliva submitted instead of sputum.
- Specimen received leaking, unsanitary, or with exterior contamination.

Samples received by the laboratory are potentially infectious. All laboratory staff are trained in handling potentially infectious substances; however, the laboratory reserves the right to reject samples received in a condition that would unduly increase staff risk of infection (e.g. broken or leaking tubes, grossly contaminated containers).

IMPORTANT NOTES FOR REQUESTERS

- VNH Laboratory has limited resources, reagents, and staff — please request only the tests that are clinically necessary. If unsure, consult the laboratory.
- Please collect test results in a timely manner from the respective ward pigeonhole.
- If a result seems abnormal or unexpected, contact the relevant section's Officer In-Charge.

8. SPECIMEN COLLECTION GUIDELINES

Before requesting a test, consider Asher's Criteria:

- Why am I ordering the test?
- What am I going to look for in the results?
- If I find it, will it affect my diagnosis?
- How will this affect the management of the case?
- Will this ultimately benefit the patient?

8.1. REQUEST FORM INFORMATION

Every request form should include:

- Patient's full name (surname and first name)
- Patient's address.
- Hospital number (where applicable).
- Gender.
- Date of birth — do not write “Ad”/“Adult” or “Ch”/“Child” in place of an age or date of birth.
- Nature of specimen and date of collection.
- Test(s) requested.
- Relevant clinical information including symptom onset.
- Signature and printed name of the requesting clinician (no other signature is accepted, to avoid unnecessary tests being requested).

8.2. LABELLING SPECIMENS

Every specimen container must be clearly labelled with:

- Patient's full name (surname and first name).
- Date of birth.
- Date and time of collection.
- Nature of the specimen (not required for an obvious blood-collection tube, but required where, for example, an “effusion” is collected into a blood container — in that case the fluid type must be written on the label).

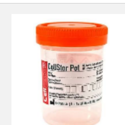
Do not label on the lid of the specimen container — write on the side.

Sample and request form information must correlate exactly. Samples are only accepted for analysis where the minimum criteria above are met; this responsibility lies with the person collecting the sample.

8.3. QUICK REFERENCE — SAMPLE TYPES & CONTAINERS

Match the sample type to the correct tube or container before collection — the wrong tube is one of the most common reasons a specimen is rejected.

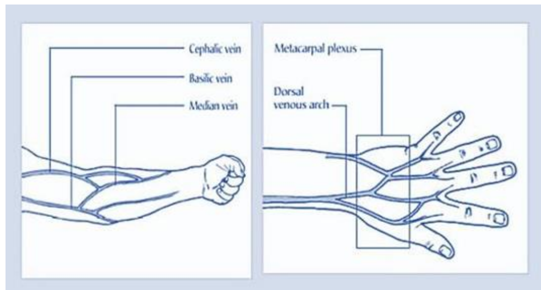
Sample type	Tube / container	
Whole blood – EDTA	Lavender top	

Sample type	Tube / container	
Plasma – Sodium Citrate	Light blue top	
Serum, no additive	Red top	
Serum, with clot activator / gel separator	Red top with Yellow ring	
Whole blood – Lithium Heparin	Green top or Royal Blue Top	
Blood culture	Aerobic/anaerobic culture bottle	
Urine	Yellow top container	
Cerebrospinal fluid (CSF)	Sterile screw-top universal container	
Stool	Brown top container	
Swab (bacterial culture)	Amies +/- charcoal transport swab	
Sputum	Yellow top container, in a biohazard bag	
Tissue biopsy	Formalin pot	
Nasopharyngeal & oropharyngeal swab (molecular/PCR)	Viral transport medium (VTM)	

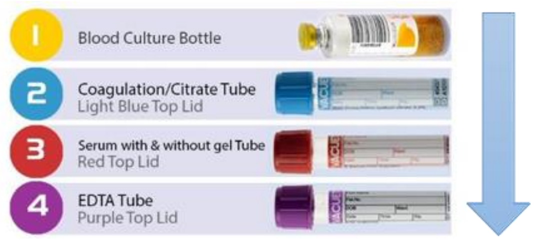
9. SPECIMEN COLLECTION PROCEDURES

9.1. VENIPUNCTURE

The image below shows the preferred sites for collection of blood;



The order of which to draw blood is shown in the diagram below;



GENERAL PROCEDURE:

a) Reassure the patient that only the minimum amount of blood required for testing will be drawn.
b) Assemble equipment appropriate to the patient's physical characteristics.
c) Wash hands and put on gloves.
d) Position the patient with the arm extended to form a straight line from shoulder to wrist.
e) Select the appropriate vein. The median cubital, basilic, and cephalic veins are most frequently used; veins become more prominent if the patient closes their fist tightly.
f) *Factors to consider in site selection:*
   - Avoid extensive scarring or healed burn areas.
   - Do not obtain specimens from the arm on the same side as a mastectomy.
   - Avoid areas of haematoma.
   - If an IV is in place, samples may be obtained below it, but never above it.
   - Do not obtain specimens from an arm with a cannula, fistula, or vascular graft.
   - Allow 10–15 minutes after a transfusion is completed before obtaining a blood sample.
g) Apply the tourniquet 3–4 inches above the collection site. Never leave it on for over 1 minute — if used only for preliminary vein selection, release and reapply after two minutes.
h) Clean the puncture site with a 70% alcohol swab in an outward spiral from the penetration point. Allow the skin to dry; do not touch the site after cleaning.

VENIPUNCTURE USING A VACUTAINER:

a) Attach the appropriate needle to the hub, twisting it tight.

b) Remove the plastic cap over the needle and hold bevel up.
c) Pull the skin tight just below the puncture site.
d) Penetrate the skin and vein in one smooth motion, in line with the vein.
e) Insert the first tube (following the proper order of draw) into the hub; blood should flow into the evacuated tube.
f) After blood starts to flow, release the tourniquet and ask the patient to open their hand.
g) When flow stops, remove the tube. If multiple tubes are needed, follow the correct order of draw to avoid cross-contamination and erroneous results.
h) Place a gauze pad over the puncture site and remove the needle. Apply pressure for at least 2 minutes; apply a fresh bandage once bleeding stops.
i) Dispose of the needle/hub into a sharps container. Label all tubes with the patient's details, initials, date, and time.

Do not shake or mix tubes vigorously. All labels on the specimen container must correlate with those on the laboratory request form.

VENIPUNCTURE USING A SYRINGE:

a) Place a sheathed needle or butterfly on the syringe and turn the bevel up.
b) Pull the skin tight just below the puncture site and penetrate the skin and vein in one motion.
c) Draw the desired amount of blood, pulling back slowly on the plunger.
d) Release the tourniquet, then apply a gauze pad and remove the needle, applying pressure for at least 2 minutes.
e) Transfer blood into the appropriate tubes as soon as possible using a needleless device. Gently invert additive tubes 5–8 times.
f) Dispose of the syringe and needle in the designated waste bin and sharps container.

VENIPUNCTURE IN INFANTS/CHILDREN:

a) Confirm the patient's identification and secure the patient in a stable position.
b) Assemble required supplies and select the collection site.
c) Proceed as for routine phlebotomy; if the child is old enough, collect as for an adult.

SPECIALISED BLOOD SAMPLE: NEONATAL BILIRUBIN SAMPLES

Container: red-top tube. Specimen: serum.

a) Following blood collection;
b) Completely wrap the specimen container in aluminium foil to protect it from light.

a. 

c) Place the wrapped container and requisition form in a biohazard carrier bag and transport to the laboratory.

9.2. BLOOD CULTURE COLLECTION

Container: blood culture bottle (aerobic and anaerobic, where adult collection is recommended).

PROCEDURE:

a) Select the appropriate vein as for routine venipuncture.
b) Apply the tourniquet and clean/sterilize the puncture site as above. Avoid re-palpating the vein after disinfecting; if necessary, apply alcohol to a gloved finger and allow it to dry first.
c) Penetrate the skin and vein with a fresh, sterile needle.
d) Inject a maximum of 10 mL of blood through the central ring of the bottle's rubber stopper.
e) Mix the blood thoroughly with the broth in the bottle.
f) Dispose of the syringe and needle appropriately
g) Label the bottle with the patient's identification details.
h) Transfer the inoculated bottle to the laboratory immediately. If transport is delayed, incubate the bottle at 36 ± 1°C and deliver within 24 hours.

One aerobic and one anaerobic bottle is recommended for adults.

9.3. STOOL COLLECTION

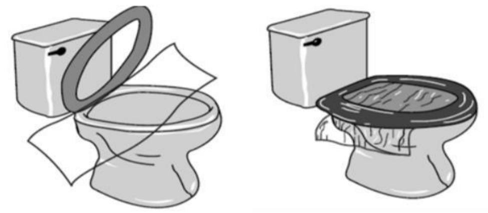
Container: Sterile Container, Brown Cap.

NOTES FOR CLINICIANS

- Collect into a clean, wide-mouthed container or collection “hat”; do not contaminate with water or urine.
- A bedpan specimen is acceptable if it contains only stool.
- Newspaper or plastic wrap may be draped over the toilet seat to assist collection.
- Do not submit diapers.
- Collect only one sample per day. Inpatients should not have a routine stool culture ordered after 3 days of admission unless food poisoning is suspected.
- Transport to the laboratory within 24 hours; refrigerate if there will be any delay.

PROCEDURE:

- Wash hands before and after collection.
- Urinate before collecting the stool so that you do not get any urine in the stool sample (female only). Do not urinate while passing the stool.
- Pass stool into a sterile cup.
- You may also tape cling or plastic wrap across the toilet to catch the stool, please see picture below.



- Either solid or liquid stool can be collected.
- If you have diarrhea, a plastic bag taped to the toilet seat may make the collection process easier; the bag is then placed in a sterile cup.
- Do not collect the sample from the toilet bowl.
- Do not mix toilet paper, water, or soap with the sample.
- Place the lid on the container and label container with your name and the date the stool was collected.
- Deliver stool specimen the same day it is collected to the laboratory.

Infants with diarrhoea are encouraged to provide a rectal swab instead.

SPECIALISED STOOL SAMPLE: ACUTE FLACCID PARALYSIS (AFP) SAMPLE FOR POLIO INVESTIGATION

a) Collect two stool samples, each about the size of an adult's thumb (or at least 5 mL if watery), at least 24 hours apart, both within 14 days of paralysis onset for the most accurate results.
b) Place in separate, clean, dry, leak-proof, tightly sealed containers.
c) Label each container with the patient's name, EPIID code, collection date, and whether it is the 1st or 2nd sample.
d) Store immediately at 4–8°C (do not freeze) and send to the VNH Laboratory Department.

9.4. URINE COLLECTION

Container: sterile container, yellow cap. Early-morning mid-stream collections are preferred; samples must reach the laboratory within 2 hours or be refrigerated and delivered within 24 hours (overnight refrigeration is acceptable). Approximately 10-20mLs is required unless otherwise stated.

PROCEDURE:

Female patients:

- Sit comfortably and clean the vaginal area front-to-back with soap, then rinse with a moistened pad, using each pad once.
- Remove the lid of the cup and hold the cup with your fingers on the outside.
- <u>First-catch urine:</u> Collect urine directly into cup – fill cup half full.
- <u>Clean-catch urine:</u> Pass a small amount of urine into the toilet first. Then move cup into urine stream and collect a mid-stream sample – fill cup half full.
- Replace and tighten the lid.

Male patients:

- Retract the foreskin if uncircumcised and clean the glans (head of penis).
- Remove the lid of the cup and hold the cup with your fingers on the outside.
- <u>First-catch urine:</u> Collect urine directly into cup – fill cup half full.
- <u>Clean-catch urine:</u> Pass a small amount of urine into the toilet first. Then move cup into urine stream and collect a mid-stream sample – fill cup half full.
- Replace and tighten the lid.

Consult the Microbiology Officer In-Charge or Senior Scientist for any queries on appropriate procedure.

SPECIALISED URINE SAMPLE: URINE FOR CYTOLOGY

Urine collection for cytology investigation requires a 2-3 hour, whole output specimen, greater than 50mLs. Early morning urines are NOT satisfactory due to advanced cellular degeneration in urine overnight. The procedure for collection is the same as above, clean-catch urine is required. Cells degenerate quickly in acid medium, therefore, all urine for cytology should immediately be mixed with an equal volume of 50% alcohol to enhance fixation. Urines must then be immediately transported to the lab.

9.5. SPUTUM COLLECTION

Sputum should be collected in a sterile plastic container. Sputum is mucous or phlegm coughed from deep in the lungs. It is not saliva. Salivary samples will be rejected.

1. Take a sterile plastic container.
2. Ensure mouth is free from foreign matter – rinse with water prior to collection.
3. Take deep breaths through your mouth and cough up mucous from deep in your lungs.
4. Open the container and hold it close to your mouth. Cough the mucous into the container.
5. 1-2 teaspoons of specimen is adequate.
6. Once collected screw lid tightly.
7. Write patient information and clinician information on specimen and bag in biohazard safety bag.
8. Deliver to laboratory as soon as possible on the same day. Refrigerate if delayed.
9. NOTE: a separate form and specimen is required for each test requested.

SPECIALISED SPUTUM SAMPLE: MYCOBACTERIUM TUBERCULOSIS INVESTIGATION

A series of at least three samples on three successive days is recommended; early-morning sputum is preferred. Should the patient be unable to provide sputum gastric aspirate can be collected instead.

PROCEDURE: GASTRIC WASH

a) Explain the procedure to the patient
b) Ensure appropriate PPE is worn.

c) Collect early in the morning, prior to eating, drinking and oral hygiene.
d) Position patient appropriately (supine or semi-recumbent)
e) Insert a nasogastric tube if not already in place.
f) Aspirate gastric contents using a sterile syringe.
g) If insufficient volume is obtained, sterile water may be instilled and re-aspirated according to clinical procedures.
h) Transfer sample into sterile container immediately and secure the lid.
i) Label specimen and transport immediately to laboratory at room temperature.
j) Do not delay transport – TB culture is time-sensitive and must be prioritised.

Urgent AFB stains will only be performed after hours following prior consultation with the microbiology scientist.

9.6. SEMEN COLLECTION

Container: sterile yellow-capped container.

NOTE FOR CLINICIANS

- Collect after a minimum of 48 hours, and no more than 7 days, of sexual abstinence.
- Semen collected in a latex condom or by coitus interruptus is not acceptable for evaluation.
- If you are unable to collect by masturbation, contact your physician to discuss an alternative collection method.
- Deliver to the laboratory within 1 hour of collection, or sperm motility may decrease and affect the result.
- Keep the sample close to the body for temperature stability — never refrigerate, unless the test is for count and morphology only (post-vasectomy).

PROCEDURE:

1. Preparation: urinate first; wash the genital area with warm water only (no soap); wash and dry hands; remove lid from specimen cup; avoid touching the inside of the container or lid.
2. Collect by masturbation directly into a sterile, wide-mouthed glass or plastic container.
3. Replace container lid immediately after.
4. Wash hands, then label the container with name, date, and time.
5. Keep sample at body temperature until given to laboratory staff.
6. Deliver to laboratory within 60 minutes in a biohazard carrier bag with the requisition form.

9.7. NASOPHARYNGEAL SWAB COLLECTION

Container: appropriately sized NP swab and viral transport medium or amies transport medium (+/-charcoal). Note VTM/UTM is used for molecular testing, amies is used for microbiological culture.

VTM/UTM NP swab

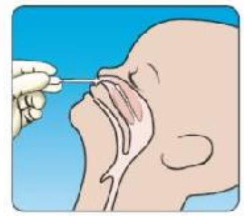
Amies swab +/- charcoal

NOTE FOR CLINICIANS

Collect within 7 days of symptom onset.

PROCEDURE:

a) Explain the procedure to the patient.
b) Wash hands and put on appropriate PPE (at minimum gloves and a face mask).
c) If there is visible nasal mucus, ask the patient to gently clear it with a tissue first — the virus is found in the cells lining the nose, not in the mucus itself.
d) Seat the patient comfortably, ideally in a high Fowler's position with the head supported; a second person may help.
e) Tilt the patient's head back slightly.
f) Advance a flexible swab gently along the floor of the nose (straight back, not upward) until reaching the posterior nasopharynx.
g) Once gentle resistance is felt, rotate the swab several times, then withdraw it.
h) Break off or seal the swab into the transport medium.
i) Remove PPE, wash hands
j) Label the specimen, and transport it with the completed requisition form.



9.8. OROPHARYNGEAL SWAB COLLECTION

Container: appropriately sized flocked swab and viral transport medium or amies transport medium (+/-charcoal). Note VTM/UTM is used for molecular testing, amies is used for microbiological culture.

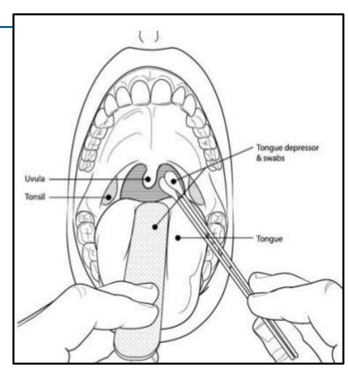
VTM/UTM OP swab

Amies swab +/- charcoal

NOTE FOR CLINICIANS

Collect within 7 days of symptom onset.

PROCEDURE:



a) Tilt the patient's head back approx. 70°.
b) Use a tongue depressor to assist visibility.
c) Advance a flexible swab to the tonsil region slowly, with a steady motion
d) Once resistance is met, rotate the swab several times and withdraw it.
e) Break off or seal the swab into the transport medium.
f) Remove PPE, wash hands.
g) Label the specimen, and transport with the completed requisition form.

9.9. CULTURE SWAB COLLECTION (ALL GENERAL BACTERIOLOGY)

Container: For any general bacterial swab culture (e.g. wound, eye, ear, nasal, through, high vaginal etc) please collect using amies/charcoal transport medium.

Amies swab +/- charcoal

PROCEDURE:

a) Explain procedure to the patient and wear appropriate PPE.
b) Expose the site to be swabbed.
c) Where pus or exudate is present, sample directly from it; otherwise moisten the swab tip first with sterile saline for a dry site.
d) Roll the swab firmly over the affected area, covering as wide a surface as possible, to collect adequate material.
e) NOTE: A dry swab with no visible material may be rejected.
f) Avoid contaminating the swab with surrounding skin, mucosa, or normal flora.
g) Return the swab to its transport medium immediately and seal (or break) as required by swab type.
h) Label the swab and request form, including the precise anatomical site i.e. left ear or wound right shin.

One swab is generally sufficient for processing on receipt; collect a second swab only if a specific additional test (e.g. a separate viral swab) is also requested. Deliver to the laboratory within 2 hours of collection.

9.10. CEREBROSPINAL FLUID (CSF) COLLECTION

Container: Sterile leak-proof screw-cap container.

PROCEDURE:

a) CSF is collected by a clinician via lumbar puncture; the laboratory does not perform the puncture itself but supplies the correct tubes on request.
b) Explain the procedure to the patient.
c) Ensure strict aseptic technique is used during invasive sampling.
d) Collect via lumbar puncture into 2–3 separate sterile screw-cap tubes, numbered/labelled (1, 2, 3) in the order they are drawn.
e) Label each tube with the patient's details, and clearly mark the collection order, since different tubes may be prioritised for different tests (e.g. the least blood-stained tube for cell count).
f) Do not refrigerate CSF intended for culture — transport at room temperature and send to the laboratory immediately.
g) Hand-deliver to the laboratory without delay; do not place CSF in a pneumatic tube system or leave it for routine collection rounds.

All CSF samples are treated as urgent on receipt, regardless of how the request form is marked.

9.11. OTHER STERILE BODY FLUIDS COLLECTION

Container: Sterile leak-proof screw-cap container. Includes pleural fluid, ascitic (peritoneal) fluid, synovial fluid, pericardial fluid, and any other normally sterile cavity aspirates.

PROCEDURE:

a) Explain the procedure to the patient.
b) Ensure strict aseptic technique is used during invasive sampling.
c) Wear appropriate PPE.
d) Collect fluid directly from sterile site using clinician-performed aspiration.
e) Ensure adequate volume is collected to allow for division into multiple test containers.
f) Label all specimens with patient identifies and precise site i.e. right pleural fluid.
g) Transport immediately to the laboratory. Do not refrigerate unless instructed to.
h) Clearly indicate suspected diagnosis (e.g. TB, sepsis) on request form.

9.12. SMEAR COLLECTION

Container: glass microscopy slides with a labelled frosted end, plus fixative (95% ethanol /Cytofix /Sprayfix). Slides and fixative are available from the laboratory storeroom.

PROCEDURE:

Use this procedure for any smear taken directly from a tissue surface — for example, an ulcerated lesion, open wound, skin lesion, or mucosal surface where scrape cytology has been requested. It is not required for smears produced from aspirated or collected fluids.

a) Assemble equipment: labelled glass slides, fixative (95% ethanol or equivalent), a clean spatula or scalpel blade, and PPE (gloves, mask where appropriate).
b) Label the frosted end of each slide clearly with the patient's name, date of birth, date of collection, and anatomical site before starting — never rely on labelling after collection.
c) Gently clean any overlying surface debris or crusting from the lesion with a dry swab if necessary, to expose fresh material beneath.
d) Using a spatula or the flat of a scalpel blade, apply firm, even pressure and scrape across the surface of the lesion in one smooth motion to collect a representative sample of cells.
e) Immediately smear the collected material thinly and evenly across the central portion of the slide in a single sweep — a thick smear will obscure cellular detail.
f) Immerse the first 1–2 slides immediately (within seconds) in 95% ethanol (or Cytofix / Sprayfix) for wet fixation. Any delay results in air-drying artefact that may hamper or prevent cytological assessment.
g) Allow any remaining thinner smears (slides 3–4 if prepared) to air-dry at room temperature; these provide a complementary preparation for certain staining techniques.
h) After a minimum of 30 minutes in fixative, wet-fixed slides may be removed, allowed to drain, and placed in a slide container for transport.
i) Complete the cytology laboratory request form fully (including site, clinical history, and date of onset if relevant) and dispatch slides with the form to the laboratory promptly.

One smear per anatomical site per slide. If multiple sites are sampled, use a separate clearly labelled slide for each. Consult the Cytology section if uncertain about fixative requirements for a specific specimen type.

SPECIALISED SMEAR SAMPLE: PAP SMEAR (CERVICAL SMEAR)

Cervical smear can be carried out by a trained cytologist, or a trained health care worker.

PROCEDURE:

a) Confirm the patient's identity and that the procedure has been explained and consented to.
b) Label the frosted end of the slide clearly with the patient's name, date of birth, date of collection, and 'Cervical smear' before beginning.
c) Position the patient in the dorsal lithotomy position and insert a bivalve speculum to expose the cervix under adequate light.
d) Using an Ayre spatula (ectocervix) or endocervical brush (endocervical canal), rotate the sampling device 360° firmly against the cervix, ensuring contact with the transformation zone.
e) Transfer the collected material immediately and evenly to the centre of the slide with a single, smooth spreading motion.
f) Fix the slide immediately (within seconds) by flooding with 95% ethanol, or by spraying with Cytofix or Sprayfix at approximately 20 cm distance. Do not allow the smear to air-dry before fixation — air-drying artefact will hamper or prevent cytological assessment.
g) Leave the slide in fixative for a minimum of 30 minutes before removing, draining, and placing in a slide transport container.
h) Complete the cytology request form with the patient's clinical details, last menstrual period, any hormone use, previous abnormal smear history, and clinical indication.

9.13. FINE NEEDLE ASPIRATION (FNA)

Container: glass microscopy slides plus fixative (95% ethanol / Cytofix / Sprayfix), sterile 21–23G needle, 10–20 mL syringe, and a syringe holder/pistol grip where available. All equipment and fixatives are prepared by the Cytology section. Contact the Cytology section before the procedure — sufficient advance notice is required for the Cytology Scientist to prepare equipment and fixatives.

The Cytology Scientist must be present at the time of collection for immediate processing of the sample; do not proceed if the Scientist is not available.

PROCEDURE:

a) Confirm the patient's identity, explain the procedure, and obtain consent. Position the patient comfortably with the target lesion accessible and well-supported.
b) Label the frosted end of each slide with the patient's name, date of birth, date of collection, and anatomical site before starting.
c) Clean the overlying skin with an alcohol swab and allow to dry. Sterile technique is required throughout.
d) Palpate and stabilise the lesion between the fingers of the non-dominant hand. For deep or non-palpable lesions, ultrasound or imaging guidance is required — coordinate with the relevant clinical team and notify the Cytology section accordingly.

e) Attach the needle to the syringe (or syringe pistol) and, with negative pressure applied, insert the needle into the lesion with a single firm movement.
f) Maintaining negative pressure, move the needle back and forth through the lesion several times using short rapid strokes (5–10 passes) across different planes to sample adequately.
g) Release the negative pressure before withdrawing the needle — failure to do so will aspirate material into the barrel of the syringe where it cannot be retrieved.
h) Withdraw the needle and apply gentle pressure to the puncture site with a gauze pad for at least 2 minutes.
i) Detach the needle, draw air into the syringe, reattach the needle, and gently expel the aspirated material onto the centre of a labelled slide.
j) The Cytology Scientist present will prepare smears immediately from the expelled material, applying wet fixation (95% ethanol or spray fixative) and/or air-drying as appropriate for the staining methods planned.
k) Any residual material in the needle hub may be rinsed into a small volume of saline or cell-preservative medium provided by the Cytology section for additional cell block preparation.
l) Complete the cytology request form fully — include the anatomical site, clinical history, size and character of the lesion, and any relevant imaging findings — and dispatch with the slides.

A minimum of 2–4 passes is generally recommended for adequate cellularity, though the number will depend on the lesion type and material obtained. If a cyst is aspirated, collect the fluid into a sterile container and hand it directly to the Cytology Scientist for processing.

9.14. TISSUE BIOPSY COLLECTION

Container: Pre-filled formalin pot. Ensure adequate container size to fully immerse specimen without compression.

NOTE FOR CLINICIANS

Histology specimens must be immediately fixed in formalin to prevent autolysis and preserve tissue architecture. Delay in fixation will significantly compromise diagnostic quality. Do not refrigerate formalin-fixed specimens.

PROCEDURE:

a) Confirm patient identification and ensure correct specimen request form is completed, including full clinical history and anatomical site of collection.
b) Explain the procedure to the patient and ensure informed consent has been obtained by the treating clinician.
c) Wear appropriate PPE (gloves, mask, and eye protection where splash risk exists).
d) Collect tissue specimen using sterile surgical technique appropriate to the procedure (e.g. biopsy, excision, endoscopy, or punch biopsy).
e) Immediately transfer the fresh tissue specimen into the pre-filled formalin container. Ensure the specimen is fully submerged.
f) Ensure the volume of formalin is at least 10 times the volume of the tissue specimen to allow adequate fixation.
g) Do not crush, squeeze, or dry the specimen prior to fixation, as this will distort histological architecture.
h) If multiple specimens are taken from different sites, place each in a separate, clearly labelled formalin pot indicating the exact anatomical site.
i) Securely close the container lid immediately to prevent leakage.

j) Complete and attach the histology request form, ensuring all relevant clinical information and suspected diagnosis is included.
k) Transport specimens to the laboratory at room temperature as soon as possible.
l) Ensure all formalin containers are placed in appropriate biohazard transport bags and handled according to chemical safety guidelines.

10. HANDLING AND TRANSPORTATION OF SPECIMENS TO VNH LABORATORY

10.1. BIOHAZARD CARRIER BAG

All specimens must be transported to the laboratory in a sealed biohazard carrier bag, with the requisition form placed in the bag's outer document pocket (not in direct contact with the specimen).

a) Tubes containing blood must be placed a biohazard specimen bag.
b) Urine specimens and other biological fluids must be placed in separate biohazard bags. For example: Do not put a tube of blood and a urine specimen in the same bag.
c) Ensure that the urine containers are tightly closed before being placed in the bags.
d) The requisition accompanying the specimen should be folded and placed in the pouch on the outside of the bag.
e) Do not put the requisition in the same compartment as the specimens.
f) The label identifying the patient must be placed directly onto the specimens and not on the bag. Specimens identified only on the bag, will be regarded as not identified and will be rejected by the laboratory.
g) Make sure the bags are tightly closed (same principle as "Ziploc" bags).

10.2. SPECIMEN DELIVERY

All specimens must be delivered in the correct specimen containers, inside biohazard bags, with sample request form completed. Sample collected during working hours should be delivered to the reception desk for reception and registration. Specimens received after hours should be delivered to the reception desk with a special note on the request form alerting the on-call scientist to the urgent or unusual nature of the specimen. A phone call ahead must be made to the on-call staff.

11. CUSTOMER FEEDBACK AND COMPLAINTS

The Medical Laboratory Services is committed to continual improvement and actively seeks feedback from its users to ensure that laboratory services continue to meet the needs and expectations of clinicians and patients.

To support this commitment, the laboratory conducts an annual customer satisfaction survey involving hospital wards, outpatient departments, and private medical clinics. The survey evaluates key aspects of laboratory performance, including the quality and reliability of test results, turnaround times, communication, specimen collection services, accessibility of laboratory staff, and overall customer satisfaction. Feedback obtained through these surveys is reviewed by laboratory management and incorporated into quality improvement activities, corrective actions, and service planning where appropriate.

In addition to the annual survey, compliments, suggestions, and complaints are welcomed at any time and are reviewed in accordance with the laboratory's Quality Management System. All feedback is documented, investigated where necessary, and used as an opportunity to improve laboratory services. Tabulated below are the nature of the complaint/feedback and who it should be directed.

Nature of Complaint	Person the complaint should be directed to
Minor (e.g. need another copy of laboratory result, delay result, blood not collected etc.)	Head of Laboratory section concerned (specific medical lab scientist)
Serious (verbal or written complaint) e.g. wrong blood bag unit issued; urgent lab investigation not carried out etc.	QC Officer or Principal Medical Laboratory Officer
Major incidents (e.g. fire in the laboratory, prolong power failure etc.)	Principal Medical Laboratory Officer

Refer to Annexure 7 for the Customer Compliment and Complaint Form.

12. DEPARTMENTAL SERVICES

The general policy below applies to every department unless stated otherwise:

- Specimen delivery: specimens collected during working hours go to laboratory reception for registration; after-hours specimens are delivered to reception with an urgency note for the on-call scientist (and a phone call ahead, where possible).
- Processing: testing is carried out under quality-controlled, standardized procedures set out in each section's Standard Operating Procedures, to ensure validated, quality-assured reports.
- Delivery of test reports: results are placed in the relevant ward pigeonhole; the collecting clinician/nurse signs the Result Register Log Book.

If you are viewing the handbook online, please Ctrl+click on the department you are interested in, to go straight there.

Department-specific test tables below give the routine and urgent turnaround time for each test.

The Flag column:

- C = CRITICAL (a defined critical-value alert and phone-notification requirement);
- S = SEND-AWAY (not processed on-site; referred overseas).

13. HAEMATOLOGY SERVICES

These activities are carried out under strict quality controlled, approved standardised procedures stated in the Haematology Standard Operating Procedures to ensure that the quality and validated results are reported/dispatched.

Services include malaria diagnostics. INR and electrolytes are treated as sensitive tests and are processed immediately regardless of the urgency.

13.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
Full Blood Count (FBC)	4mL Whole blood – EDTA	1 hour	15-30 min	
White Blood Cell Differential Count	Whole blood – EDTA	1 hour	30 min	
Reticulocyte Count *Reference range – Adults 0.2-2%; Infant 2-6%*	4mL Whole blood – EDTA	4 hours	1 hour	
Peripheral Blood Film Examination **A blood film is not routinely done on FBC requests unless specifically requested for. We are trying to do films on all abnormal FBC results as a routine but only when special trained* *blood films expertise scientist are available.*	4mL Whole blood – EDTA	24 hours	2 hours	
Erythrocyte Sedimentation Rate (ESR) *Reference range: Male 0-15mm/hr; Female 0-20mm/hr*	4mL Whole blood – EDTA	2 hours	1.5 hour	
White Blood Cell Count – Manual	4mL Whole blood – EDTA	1 hour	30 min	
Prothrombin Time (PT) & INR – Automated *Normal range 12-16sec* *Therapeutic range: 2.5-4*	4.5mL Plasma – Sodium Citrate	1 hour	30 min	C
International Normalised Ratio (INR) – Manual *Therapeutic range: 2-4*	4.5mL Plasma – Sodium Citrate	1 hour	30 min	C
Malaria Rapid Diagnostic Test (RDT)	4mL Whole blood – EDTA	1 hour	30 min	
Malaria Microscopy	4mL Whole blood – EDTA	2 hours	1 hour	

Test	Sample	Routine TAT	Urgent TAT	Flag
Microfilariae Parasite Screening Microscopy *This specimen should be collected at night-time after 2200hrs to 0400hrs (Peak 2400hrs)*	4mL Whole blood – EDTA	2 hours	NA	
Phlebotomy (Bleeding Time)	At bedside	30 min	15 min	

13.2. CRITICAL VALUES (PHONED TO REQUESTING CLINICIAN ONCE VERIFIED):

Test	Critical low	Critical high
Prothrombin Time	—	> 17 sec
Haemoglobin (0–7 weeks)	≤ 8.0 g/Dl	≥ 24.0 g/Dl
Haemoglobin (> 7 weeks / adult)	≤ 8.0 g/Dl	≥ 20.0 g/Dl
INR	—	≥ 5.0
Leukocytes	—	≥ 100.0 $\times 10^9$/L
Absolute Neutrophil Count / Neutrophils	≤ 0.5 $\times 10^9$/L	—
Platelets	≤ 40 $\times 10^9$/L	≥ 1000 $\times 10^9$/L
CSF White Blood Cell Count	—	≥ 100.0 cells/Ml

13.3. DELIVERY OF TEST RESULTS

Haematology laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

14. BLOOD BANK / TRANSFUSION SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Blood Transfusion Standard Operating Procedures to ensure that the blood units dispensed are safe and appropriate for each specific patient's need. Includes donor recruitment. Cryoprecipitate has the longest lead time of all blood products — plan ahead.

14.1. SPECIAL REQUIREMENTS FOR SAMPLE COLLECTION

- The tube must be accurately and clearly labelled in ink at the bedside, with the patient positively identified.
- All details on the Blood Transfusion Request Form must be completed and signed by the clinician.
- Relevant clinical history (previous pregnancy, haemolysis-causing drugs, previous transfusions) should be included on the form.

14.2. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
ABO and Rh(D) Blood Grouping	4mL Whole Blood – EDTA or Serum Red top	30 min	15 min	
Crossmatch Compatibility Testing	4mL Whole Blood – EDTA or Serum Red top	2 hours	1 hour	
Coomb's Test (DCT/ICT)	4mL Whole Blood – EDTA or Serum Red top	1 hour	30 min	
Indirect Antiglobulin Test (IAT)	4mL Whole Blood – EDTA or Serum Red top	1 hour	30 min	
Packed Red Blood Cell (PRBC) Prep & Issue	N/A	2 hours	1 hour	
Fresh Whole Blood Prep & Issue	N/A	2 hours	1 hour	
Fresh Frozen Plasma (FFP) Prep & Issue	N/A	2 hours	1 hour	
Platelet Concentrate Prep & Issue	N/A	4 hours	2 hours	
Platelet-Rich Plasma (PRP) Prep & Issue	N/A	4 hours	2 hours	
Cryoprecipitate Prep & Issue	N/A	24 hours	24 hours	
Blood Donor Recruitment & Assessment	N/A	20 min	15 min	
Blood Donor Collection	N/A	30 min	15 min	

14.3. CLINICAL USE OF BLOOD

Transfusion should only be used to treat a condition causing significant morbidity or mortality that cannot be managed effectively by other means. The need for transfusion can often be reduced by early treatment of anaemia, correction of iron stores before planned surgery, use of IV replacement fluids, and good surgical/anaesthetic management. Blood products carry risk of transfusion-transmissible infection (HIV, hepatitis B/C, syphilis, malaria, Chagas disease) and of acute or delayed reactions — see Annexure 1. Blood donated by family/replacement donors carries a higher risk of transfusion-transmissible infections than blood donated by voluntary non-remunerated donors. Paid blood donors generally have the highest incidence and prevalence of transfusion-transmissible infections. Blood should not be transfused unless it has been obtained from appropriately selected donors, has been screened for transfusion-transmissible infections and tested for compatibility between the donor's red cells and the antibodies in the patient's plasma, in accordance with national requirements.

14.4. SETTING UP AND MONITORING AN INFUSION

- Re-check compatibility labels against the recipient's identification immediately before infusing — there must be no discrepancy. Record the re-check with date/time.
- Use an administration set with a filter, changed at least every 24 hours or after 4–6 units. Flush the line with a blood-compatible solution before and after infusion. Do not add IV medication directly to blood.
- Most units are transfused over 1½–2 hours in adults with symptomatic anaemia, and should not exceed 4 hours, to limit bacterial proliferation and haemolysis risk at room temperature.
- A unit not used in the ward must be returned to the Blood Bank within 30 minutes of issue; blood must never be left at room temperature or in a ward fridge for more than 30 minutes.
- Observe the patient closely in the first 5–10 minutes (when most life-threatening reactions occur) and continue observation throughout and after the transfusion. Report suspected reactions immediately, or within 24 hours, to the Blood Bank and the responsible doctor.

14.5. AVAILABLE BLOOD PRODUCTS

Product	Volume	Indication
Whole Blood	300–450mL	Major transfusions, to increase red cell mass and blood volume
Packed Red Blood Cell (PRBC)	250–350mL	Moderate to severe symptomatic anaemia
Fresh Frozen Plasma (FFP)	200–300mL	Coagulation factor deficiency (e.g. warfarin overdose, DIC); not a plasma volume expander
Platelet Concentrate	50–60mL	Bleeding due to thrombocytopenia; usually 6 adult units, or 1 unit per 10–12 kg in children
Cryoprecipitate	30 – 40mL	Treatment of prolong PT or APTT, volume replacement, Haemophilia A AND massive haemorrhage with low fibrinogen

14.6. RISKS OF TRANSFUSION

In some clinical situations, transfusion may be the only way to save life or rapidly improve a serious condition. However, before prescribing blood or blood products for a patient, it is always essential to weigh up the risks of transfusion against the risks of not transfusing.

14.7. RED CELL TRANSFUSION

a) The transfusion of red cell products carries a risk of serious haemolytic transfusion reactions.
b) Blood products can transmit infectious agents, including HIV, hepatitis B, hepatitis C, syphilis, malaria and Chagas disease to the recipient.
c) Any blood product can become contaminated with bacteria and very dangerous if it is manufactured or stored incorrectly.

14.8. PLASMA TRANSFUSION

a) Plasma can transmit most of the infections present in whole blood.
b) Plasma can also cause transfusion reactions.
c) There are few clear clinical indications for plasma transfusion. The risks very often outweigh any possible benefit to the patient.

14.9. BLOOD SAFETY

The quality and safety of all blood and blood products must be assured throughout the process from the selection of blood donors through to their administration to the patient. This requires:

a) The establishment of a well-organized blood transfusion service with quality systems in all areas.
b) The collection of blood only from voluntary non-remunerated donors from low-risk populations and rigorous procedures for donor selection.
c) The screening of all donated blood for transfusion-transmissible infections: HIV, hepatitis viruses, syphilis and, where appropriate, other infectious agents, such as Chagas disease
d) and malaria.
e) Good laboratory practice in all aspects of blood grouping, compatibility testing, component preparation and the storage and transportation of blood and blood products.
f) A reduction in unnecessary transfusions through the appropriate clinical use of blood and blood products, and the use of simple alternatives to transfusion, wherever possible.

14.10. BLOOD GROUP COMPATIBILITY

For safe transfusion practice compatible blood must be given. When a transfusion is given, it is preferable that patients receive blood of the same ABO and Rh (D) group, however in an emergency, if the required red cell blood group is unavailable, a patient may be given other red cell groups as shown below.

Red cell compatibility chart:

		DONOR			
	RHD	O +	A +	B +	AB +
RECIPIENT	O +	🩸			
	A +	🩸	🩸		
	B +	🩸		🩸	
	AB +	🩸	🩸	🩸	🩸

		DONOR			
	RHD	O -	A -	B -	AB -
RECIPIENT	O -	🩸			
	O +	🩸			
	A -	🩸	🩸		
	A +	🩸	🩸		
	B -	🩸		🩸	
	B +	🩸		🩸	
	AB -	🩸	🩸	🩸	🩸
	AB +	🩸	🩸	🩸	🩸

Please see Annexure 1 for details about blood transfusion adverse reactions.

14.11. BLOOD DONATION

Blood donation is a vital component of healthcare and plays an essential role in saving lives. A safe and adequate supply of donated blood ensures that hospitals can provide timely treatment for patients experiencing medical emergencies, undergoing surgery, or requiring ongoing transfusion therapy.

VNH Hospital Laboratory Blood Bank Department Accept routine blood donors in week days, emergency donors during weekends and conducts mobile blood drive to business houses and NGO.

14.12. DONOR ACCEPTANCE CRITERIA

A donor may be accepted for blood donation if all of the following criteria are met:

- Age:
  - Donor is between **18 and 60 years** of age - donors outside of this age range will not be accepted
  - Body Weight - minimum body weight of **50 kg** for a standard 450 mL whole blood donation.
- General Health:
  - Good general health
  - Well on day of donation
  - Free from fever or signs of acute illness
  - No active infection requiring medical treatment
- Haemoglobin Level

    - Female donors: **≥12.5 g/dL**
    - Male donors: **≥13.5 g/dL**
    - Haemoglobin shall be assessed before donation using an approved screening method.
- Vital Signs
    - Blood pressure: **100–180 mmHg systolic** and **50–100 mmHg diastolic**
    - Pulse: **50–100 beats per minute**, regular
    - Body temperature: **≤37.5°C**
- Medical History
    - Complete the donor health questionnaire
    - Pass the confidential medical interview
    - Have no medical condition that may endanger the donor or recipient
    - Not be taking medications that are considered unsuitable for blood donation
- Infectious Disease Risk Assessment
    - No history or current evidence of blood-borne infectious diseases.
    - No high-risk behaviours associated with HIV, Hepatitis B, Hepatitis C, or other transfusion-transmissible infections.
    - No recent exposure requiring temporary deferral.
    - Temporary deferral will apply to donors traveling to an area experiencing outbreaks of diseases that may be transmitted through blood (e.g., HIV, Malaria, dengue, Zika virus, chikungunya, or other emerging infections)
- Pregnancy and Breastfeeding
    - Pregnant women shall not donate blood
    - Breastfeeding mothers should only donate after the recommended postpartum period in accordance with national guidelines
- Donation Interval
    - Minimum interval between donations:
        - **Male donors:** At least **12 weeks (3 months)**
        - **Female donors:** At least **16 weeks (4 months)**
- Alcohol and Drug Use
    - The donor shall not be under the influence of alcohol or recreational drugs.
    - Alcohol should not have been consumed within the previous **24 hours**, where required by local policy.
- Identification
    - Provide valid identification
    - Be correctly registered in the Blood Bank information system
    - Give informed consent before donation

14.13. BLOOD UNIT ACCEPTANCE

A donated blood unit shall be accepted only when:

- Donor eligibility criteria have been satisfied.
- The blood collection process was completed without significant complications.
- The collection bag contains the required blood volume.

- All donor records and labels are complete and accurate.
- Samples for mandatory infectious disease screening have been collected.
- The blood unit passes all required quality checks before release for processing or transfusion.

14.14. REASONS FOR DEFERRAL

A donor shall be deferred if any of the following apply:

- Low haemoglobin.
- Fever or acute illness.
- Pregnancy.
- High-risk exposure to infectious diseases.
- Recent surgery or blood transfusion.
- Recent vaccination (where applicable).
- Certain chronic medical conditions.
- Medication affecting donor or recipient safety.
- Failure to meet minimum age or weight requirements.
- Any condition identified by the Medical Officer as unsuitable for donation.

14.15. BLOOD TRANSFUSION – WHEN A NEW SAMPLE IS REQUIRED

A new pre-transfusion sample shall be requested when:

- The existing sample is more than **72 hours old** in recently transfused or pregnant patients.
- Patient identification details are incomplete, incorrect, or inconsistent.
- The specimen is haemolysed, clotted (where anticoagulated blood is required), contaminated, or of insufficient volume.
- Additional testing cannot be completed on the original specimen.
- The laboratory has reason to question the integrity or identity of the sample.

A post-transfusion blood sample shall be collected when:

- A transfusion reaction is suspected.
- The clinician requests assessment of the effectiveness of the transfusion (e.g., post-transfusion hemoglobin or platelet count).
- Additional compatibility testing or serological investigations are required.
- Laboratory policy requires confirmation of blood group or antibody status.

14.16. DELIVERY OF TEST BLOOD RESULTS

Blood Transfusion laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

15. SEROLOGY SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Serology Standard Operating Procedures to ensure that the quality and validated results are reported/dispatched.

Most serology tests run on a 24-hour routine cycle; urgent requests can be escalated to 2 hours. Viral load testing is performed in-house unless out of stock.

15.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
HIV-1/2 Antibody Screening	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	
HIV-1/2 Confirmatory Testing	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	
HIV-1 Viral Load	4mL Whole Blood - EDTA	24 hours	2 hours	
CD4 T-Lymphocyte Count	4mL Whole Blood - EDTA	24 hours	2 hours	
Hepatitis B Surface Antigen (HBsAg)	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	
Hepatitis B Viral Load	4mL Whole Blood - EDTA	24 hours	2 hours	
Hepatitis C Antibody Test	Serum, plasma or whole blood	24 hours	2 hours	
Syphilis Rapid Plasma Reagin (RPR)	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	
Syphilis Treponema pallidum (TP) Rapid Test	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	
Dengue RDT (NS1 Antigen and/or IgM/IgG)	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	
Leptospirosis Rapid Diagnostic Test	4mL Whole Blood – EDTA or Serum Red Top	24 hours	2 hours	

Test	Sample	Routine TAT	Urgent TAT	Flag
Leptospirosis Confirmation / Serotyping	4mL Serum	Weeks	Weeks	S
Dengue Serotyping	4mL Serum	Weeks	Weeks	S
Autoimmune Antibody Testing (SLE, ANA, anti-dsDNA)	4mL Serum	Weeks	Weeks	S
Membrane Antibody Testing	4mL Serum	Weeks	Weeks	S
Measles / Rubella Serology (ELISA)	4mL Serum	24 hours	2 hours	
Mumps / Chickenpox Serology	4mL Serum	Weeks	Weeks	S
Retroviral Studies (beyond HIV)	4mL Serum	Weeks	Weeks	S

15.2. DELIVERY OF TEST RESULTS

Serology laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

16. BIOCHEMISTRY SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Biochemistry Standard Operating Procedures to ensure that the reports dispensed are valid and of quality.

Core chemistry runs on the Cobas C311 / Fujifilm analyser; immunoassay testing runs on the E411 analyser. Available during normal working hours and after hours.

16.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
Electrolytes & Minerals (Sodium, Na/Potassium, K/Chloride, Cl/Calcium, Ca/Magnesium, Mg/Phosphate, PO4)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	C
Bicarbonate (CO_2)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	C
Renal Function Tests (RFT/Urea/Creatinine, Cr/Estimated Glomerular Filtration Rate, eGFR/Uric Acid, UA)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	
Liver Function Tests (LFT/Albumin, Alb/Bilirubin, Bil/Alanine Aminotransferase, ALT/Aspartate Aminotransferase, AST/Alkaline Phosphatase ALP/Total Protein, TP/Gamma-Glutamyl Transferase, GGT)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	C
Lipid Profile (Total Cholesterol, TC/High-Density Lipoprotein Cholesterol, HDL/Low-Density Lipoprotein Cholesterol, LDL/ Triglycerides, TG)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	
Glucose	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	C
Fasting Glucose	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	C
Glycated Haemoglobin, HbA1c	4mL Whole Blood - EDTA	2–3 hours	< 1 hour	
Cardiac & Muscle Enzymes	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	

Test	Sample	Routine TAT	Urgent TAT	Flag
(Creatine Kinase, CK/ Creatine Kinase-MB, CK-MB/Lactate Dehydrogenase, LDH/Amylase, AMY)				
Inflammation Markers (C-reactive Protein, CRP/Antistreptolysin O, ASO/Rheumatoid Factor, RF-II/Complement Component 3/4, C3/C4/ Transferrin, TRF)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	
D-Dimer	4mL Serum with activator or Serum	2–3 hours	< 1 hour	
Thyroid Function Tests (Thyroid Stimulating Hormone, TSH/Triiodothyronine, T3/Thyroxine, T4)	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	
Cardiac Troponin	4mL Serum with activator or Serum or Whole blood - Heparin	2–3 hours	< 1 hour	C
CSF & Sterile Body Fluid Chemistry	CSF / sterile fluids	2–3 hours	< 1 hour	
Urine Creatinine	Urine; MID-STREAM	2–3 hours	< 1 hour	
Tumour Markers (Prostate Specific Antigen, PSA/Alpha-Fetoprotein, AFP/Carcinoembryonic Antigen, CEA/Cancer antigen 125, CA125/Cancer Antigen 19-9, CA19-9)	4mL Serum with activator or Serum	2–3 hours	< 1 hour	
Pregnancy Marker βhCG	4mL Serum with activator or Serum	2–3 hours	< 1 hour	
Vitamin B12	4mL Serum with activator or Serum	2–3 hours	< 1 hour	
Specialist Iron Studies	4mL Serum with activator or Serum	Weeks	Weeks	S

16.2. CRITICAL VALUES (PHONED TO REQUESTING CLINICIAN ONCE VERIFIED):

Test	Critical low	Critical high
Bilirubin Total, Serum (< 1 yr)	—	≥ 255 µmol/L
Calcium, Total	≤ 1.625 mmol/L	≥ 3.25 mmol/L
Creatinine (1 day–4 weeks old)	—	≥ 132 µmol/L
Creatinine (5 weeks –23 months old)	—	≥ 176 µmol/L

Test	Critical low	Critical high
Creatinine (2yrs-11 yrs)	—	≥ 221 µmol/L
Creatinine (11yrs- 15 yrs)	—	≥ 265 µmol/L
Creatinine (≥ 16 yrs)	—	≥ 884 µmol/L
Creatine Kinase, Total	—	≥ 10,000 U/L
Glucose (< 4 weeks)	≤ 2.2 mmol/L	≥ 22.2 mmol/L
Glucose (≥ 4 weeks)	≤ 2.7 mmol/L	≥ 22.2 mmol/L
Magnesium, Serum	≤ 0.40 mmol/L	≥ 2.0 mmol/L
Osmolality	≤ 190 mOsm/kg	≥ 390 mOsm/kg
Potassium	≤ 2.5 mmol/L	≥ 6.0 mmol/L
Sodium	≤ 120 mmol/L	≥ 160 mmol/L

16.3. DELIVERY OF TEST RESULTS

Clinical Chemistry routine laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided. Final test report during normal working hours for routine testing will be distributed to the lab result box twice daily. One at 12.00pm and at 16.00 pm.

17. MICROBIOLOGY SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Microbiology Standard Operating Procedures to ensure that the reports dispensed are valid and of quality.

Microbiology services are only available within normal working hours, unless they meet exceptional circumstance. Exceptional circumstance includes, all CSF specimens, urgent specimens for admission to the ward, or urgent specimens for critical patient management. Blood cultures need a minimum 5-day incubation before a negative result can be confirmed. Many cultures have no separate urgent pathway — a Gram stain or microscopy result can usually be escalated while the culture continues to incubate.

17.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
Blood Culture	Blood culture	5 days	—	
CSF Cell Count	CSF	30 min	30 min	
CSF Differential	CSF	2 hours	1 hour	
CSF Gram Stain	CSF	2 hours	1 hour	
CSF Culture	CSF	2–3 days	—	
Stool Microscopy	Stool	2 hours	30 min	
Stool Culture	Stool	2–3 days	—	
Faecal Occult Blood	Stool	1 hour	30 min	
Body Fluid Cell Count & Differential	Sterile body fluids	2 hours	1 hour	
Body Fluid Gram Stain	Sterile body fluids	2 hours	1 hour	
Body Fluid Culture	Sterile body fluids	2–3 days	—	
Throat / Nasal Swab Culture	Culture swab	2–3 days	—	
Sputum Culture	Sputum	2–3 days	—	
Eye / Ear / Nasal Swab Culture	Culture swab	2–3 days	—	
Genital Swab	Culture swab	2–3 days	—	
High Vaginal Swab	Culture swab	2-3 days	—	
Pus / Wound Swab Culture	Culture swab	2–3 days	—	

Test	Sample	Routine TAT	Urgent TAT	Flag
Urine Dipstick	Urine; MID-STREAM	1 hour	30 min	
Urine Microscopy	Urine; MID-STREAM	1 hour	30 min	
Urine Microscopy, Culture & Sensitivity	Urine; MID-STREAM	2–3 days	—	
KOH Skin Scraping	Skin scraping	2 days	—	
Semen Analysis	Semen	1 day	1 day	
Pregnancy Test (β-hCG RDT)	Urine or 4mL serum	1 hour	30 min	
Rotavirus Rapid Diagnostic Test	Stool	1 hour	30 min	
Clostridium Species Identification	Isolate / specimen	Weeks	Weeks	S
Unusual / Atypical Organism Identification	Isolate / specimen	Weeks	Weeks	S

CSF is the only specimen processed without question after hours. Other urgent specimens are processed only if the patient is to be admitted, or where the clinician feels the result will significantly affect management. Consultation with the on-call scientist is advised.

17.2. DELIVERY OF TEST RESULTS

Microbiology laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

18. TUBERCULOSIS (TB) SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Tuberculosis Laboratory Standard Operating Procedures to ensure that the reports dispensed are valid and of quality.

GeneXpert gives same-shift turnaround; smear microscopy and CSF/pleural fluid AFB work run on a 24-hour cycle.

18.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
GeneXpert MTB/RIF Ultra	Sputum / CSF / sterile body fluids/ tissue	2 hours	1 hr 25 min	
Acid-Fast Bacilli (AFB) Microscopy (For TB workup, and Test of Cure 2-6 months post-treatment)	Sputum / CSF / sterile body fluids/ tissue	24 hours	—	
Pleural Fluid TB Investigation	Pleural aspirate	24 hours	—	
TB Culture (Mycobacterial Confirmation)	Sputum	Weeks	Weeks	S
Pertussis / Diphtheria Confirmation	Culture swab – NP/OP	Weeks	Weeks	S

18.2. DELIVERY OF TEST RESULTS

TB laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

19. MOLECULAR LABORATORY

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Molecular Laboratory Standard Operating Procedures to ensure that the reports dispensed are valid and of quality. Please request molecular tests using the Surveillance Laboratory request form.

GeneXpert panels give same-shift turnaround. Open RT-PCR tests run in batches — completion time depends on batch size, with urgent samples fast-tracked.

19.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
GeneXpert Respiratory Panel (SARS-CoV-2 / Flu A/B / RSV)	NP swab VTM	2–3 hours	1–2 hours	
GeneXpert Chlamydia / Gonorrhoea	Urine; **First-catch** ½ full jar	2–3 hours	1–2 hours	
Bordetella pertussis PCR	NP swab VTM	8–24 hours	4–6 hours	
Leptospirosis PCR	Urine; **Mid-stream** ½ full jar (>7d post-onset) / 4mL Whole blood – EDTA (<7d post-onset)	8–24 hours	4–6 hours	
Dengue / Zika / Chikungunya PCR Panel	4mL Whole blood - EDTA	8–24 hours	4–6 hours	
Malaria PCR	4mL Whole blood - EDTA	8–24 hours	4–6 hours	
Measles / Rubella PCR Panel	NP/OP swab VTM and 4mL Whole blood - EDTA	8–24 hours	4–6 hours	
Monkeypox PCR	Swab VTM/UTM (lesion)	8–24 hours	4–6 hours	S
SARS-CoV-2 Genomic Sequencing / Variant Typing	NP swab VTM	Weeks	Weeks	S
Influenza / RSV Confirmatory Testing	NP swab VTM	Weeks	Weeks	S

19.2. DELIVERY OF TEST RESULTS

Molecular laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

20. HISTOLOGY SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Histology Standard Operating Procedures to ensure that the reports dispensed are valid and of quality.

Currently, VNH Histology department can collect and prepare biopsies for investigation. Prepared specimens are shipped overseas to certified laboratories for investigation and confirmation. Local pathology advice is available, with overseas pathologist input for clarification.

20.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
Biopsy – Malignant Diagnosis	Tissue biopsy	3–4 weeks	3–4 weeks	S
Biopsy – In Situ Carcinoma	Tissue biopsy	3–4 weeks	3–4 weeks	S
Biopsy – Margin Assessment	Tissue biopsy	3–4 weeks	3–4 weeks	S

20.2. DELIVERY OF TEST RESULTS

Histology laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

21. CYTOLOGY SERVICES

These activities are carried out under strict quality controlled, approved standardized procedures stated in the Histology Standard Operating Procedures to ensure that the reports dispensed are valid and of quality.

Non-malignant results are issued once complete. Abnormal smears receive a verbal preliminary diagnosis locally, then confirmation overseas (NZ/Australia), adding 2–3 weeks.

21.1. TEST DIRECTORY

Test	Sample	Routine TAT	Urgent TAT	Flag
Cerebrospinal Fluid Cytology	CSF	24 hrs	1 hr	C
Fine Needle Aspiration Cytology • Salivary glands • Thyroid glands • Breast • Lymph Node	Aspirate	2 weeks	24 - 48 hrs	
Serous Fluid Cytology • Pericardial • Peritoneal • Pleural	Serous fluids	2 weeks	24 - 48 hrs	
Synovial Fluid Cytology	Synovial fluid	2 weeks	24 - 48 hrs	
Vitreous Fluid Cytology	Vitreous fluid	2 weeks	24 - 48 hrs	
Nipple Discharge Cytology	Discharge	2 weeks	24 - 48 hrs	
Lung Cytology • Sputum • Bronchoalveolar lavage (BAL) • Bronchial brushing/washing • Endobronchial ultrasound and guided transbronchial needle aspiration (EBUS-TBNA)	Respiratory samples	2 weeks	24 - 48 hrs	
Gastric Cytology • Oesophageal brushing/washing • Gastric lavage	GI samples	2 weeks	24 - 48 hrs	
Urine Cytology	Urine	2 weeks	48 hours	
Gynaecological Cytology (Pap smear)	Cervical smear	1 month	1 week	
Reproductive System Cytology	Reproductive tract samples	2 weeks	24 - 48 hrs	

Test	Sample	Routine TAT	Urgent TAT	Flag
Liver Cytology	Bile Duct Brushing and Biliary Duct Brushing	2 weeks	24 - 48 hrs	
Renal Cytology	Renal Brushing or washing	2 weeks	24 - 48 hrs	
Scraping and Miscellaneous Smears	Skin and Cutaneous Lesions, Cervical & Vaginal Scrapings, Oral & Buccal Mucosa Scrapings, Ocular & Conjunctival Scrapings	2 weeks	24 - 48 hrs	
Hydrocoele Cytology	Hydrocoele fluid	2 weeks	24 - 48 hrs	

Cervical smears must be fixed within seconds of collection (95% ethanol, Cytofix, or Sprayfix) to avoid air-drying artefact. FNA samples require the Cytology Scientist present at collection — notify the section in advance. See Specimen Collection Procedures for more details.

21.2. DELIVERY OF TEST RESULTS

Cytology laboratory reports will be placed in each respective wards' pigeonhole. Any clinician picking up the results at the laboratory reception must enter their name, signature and collection time on the book provided.

22. OTHER / SPECIALIST REFERRED TESTS

These have no dedicated local department and are referred out as standard practice.

Test	Sample	Routine TAT	Urgent TAT	Flag
Paternity / DNA Testing	EDTA whole blood	Weeks	Weeks	S
Any other test not held at VNH	As advised by department	Weeks	Weeks	S

23. LABORATORY-BASED SURVEILLANCE

The VNH Laboratory Department works along with Ministry of Health through the Public Health Surveillance Unit to provide confirmation of early warning of outbreaks (core syndromes and vaccine preventable diseases) so that immediate action can be taken.

All specimens collected for surveillance must be accompanied by a fully completed Laboratory-Based Surveillance Form (see Annexure 5). Forms must be completed thoroughly; the date of onset of illness must always be included. Specimens collected during working hours go to reception for registration; after-hours specimens go to reception with an urgency note and, ideally, a phone call ahead.

23.1. CORE SYNDROMES

Four core syndromes are to be reported on a weekly basis by surveillance focal point in all sentinel sites. An additional syndrome was included in 2016 in the context of the Zika epidemic and is named Dengue-Zika-Chik like illness (DZCLI). In addition, any unexpected events, deaths, cluster of illness, bird or animal die off should be reported by sentinel sites.

Syndrome	Case definition	Samples to collect
Acute Fever and Rash (AFR); *Suspected cause: Measles, Dengue virus, Zika virus, Leptospirosis, Chikungunya virus, Rubella, coxsackie*	Fever (reported or measured >= 38C) plus non-blistering rash.	• 4 mL blood, red top — serology (measles, dengue, leptospirosis, chikungunya) OR dried blood spot (DBS) for referral serology/PCR — collect at first contact or within 28 days of rash onset • Blood culture if fever ≥38°C • NP or OP swab (viral transport medium) — within 5–7 days of rash onset, from the first cases in a cluster
Prolonged Fever (PF) *Suspected cause: Typhoid, Dengue virus, Zika virus, Chikungunya virus, Leptospirosis, Malaria, pneumonia, secondary bacterial infection*	Any fever (reported or measured >= 38C) lasting three or more days.	• Blood culture if fever >38°C • 4 mL blood, red top — rapid tests for dengue and/or leptospirosis (paired sample ideal) • DBS — for referral serology/PCR (dengue, Zika, leptospirosis, chikungunya) • EDTA whole blood (4 mL) — for in-house Molecular Laboratory PCR (Section 11.7) where available • Stool (5–10 g, sterile screw-top container) — if typhoid suspected
Influenza-Like Illness (ILI) *Suspected cause: Influenza viruses,*	Fever (>= 38C) plus cough and/or sore throat.	• NP or OP swab (viral transport medium) — for influenza and respiratory virus testing • Sputum (sterile container) — if pneumonia suspected, for culture

Syndrome	Case definition	Samples to collect
respiratory syncytial virus (RSV), other resp viruses, streptococcus pneumoniae, other bacterial pneumonias		
Watery Diarrhoea (WD) *Suspected cause: Staphylococcus aureus toxin, Bacillus cereus toxin, Vibrio cholera, Clostridium perfringens toxin, Norovirus, Adenovirus, Rotavirus, Salmonella*	Three or more loose or watery stools in 24 hours.	• Stool — 5–10 g fresh stool in a sterile screw-top container, delivered within 2 hours. Collect if: bloody diarrhoea, fever >2 days, severe watery diarrhoea or dehydration, or a cluster is identified • RDT for rotavirus/adenovirus if available
Dengue/Zika/Chikungunya-Like Illness (DZCLI) *Suspected cause: Dengue virus, Zika virus, Chikungunya virus.*	Fever for at least 2 days plus two or more of: nausea/vomiting, muscle or joint pain, severe headache or retro-orbital pain, rash, spontaneous bleeding (suspect dengue), or conjunctivitis (suspect Zika).	• 4 mL blood, red top — dengue NS1 antigen and/or IgM/IgG RDT, or DBS for referral serology/PCR • EDTA whole blood (4 mL) — for in-house Molecular Laboratory PCR panel. • Urine (sterile container, ≥5 mL) — for Zika PCR, up to 14 days after symptom onset

DBS (dried blood spot, filter paper — fill all 3 circles) is used for referral serology and/or PCR at a reference laboratory. It is particularly practical where transporting a liquid serum sample is not feasible (e.g. outer islands). It is not used for in-house RDT, which requires fresh whole blood or serum directly on the test strip.

23.2. VACCINE-PREVENTABLE DISEASE (VPD) / HOSPITAL-BASED ACTIVE SURVEILLANCE (HBAS)

The focus of VPD surveillance is to monitor and assess the impact of strategies and activities for reducing morbidity and mortality from vaccine-preventable diseases. Collection, analysis, and interpretation of surveillance data is vital to guide vaccination policies and programs, and to ensure immunisation targets are being reached.

The following diseases are under active surveillance:

Syndrome	Case definition	Samples to collect
Acute Flaccid Paralysis (AFP)	Any case of acute flaccid paralysis in a child under 15 years of age.	• Two stool samples, each approximately thumb-sized (≥5 mL if watery), collected at least 24 hours apart, both within 14 days of paralysis onset — earlier collection gives the most reliable result • Place each sample in a separate clean, dry, leak-proof, tightly sealed container • Label each with the patient's name, EPIID code, collection date, and whether it is sample 1 or 2 • Store immediately at 4–8°C (do not freeze) and send to VNH Laboratory without delay • Also collect 4 mL blood, red top, for serology if Guillain-Barré / Zika is suspected
Neonatal Tetanus (NT)	A neonate with a normal ability to suck or cry during the first two days of life, which between days 3 and 28 becomes unable to suck or cry normally, and becomes stiff or develops jerking movements or convulsions, or a combination of these	• Neonatal tetanus is a clinical diagnosis — there is no specific laboratory test to confirm it • Collect 4 mL blood, red top — for blood culture (to exclude sepsis as an alternative diagnosis) and baseline bloods • Wound swab (Amies transport medium) from the umbilical stump or any visible wound — for Clostridium tetani culture (note: culture is often negative and a negative result does not exclude the diagnosis) • Document maternal vaccination history and delivery conditions on the surveillance form
Diphtheria	A person with an illness of the upper respiratory tract characterised by laryngitis, pharyngitis, or tonsillitis, and adherent membranes on the tonsils, pharynx, and/or nose	• Throat swab AND nasal swab (Amies transport medium with or without charcoal) — collect before antibiotics are started where at all possible, using the oropharyngeal swab procedure (Section 9.12); attempt to sample material from beneath or at the edge of the membrane • A second paired set of swabs is recommended if the first was collected after antibiotics were started • Validate DPT/pentavalent/Td/DT vaccination history and record on the Case Investigation Form (CIF)

Syndrome	Case definition	Samples to collect
		• Confirmation (Corynebacterium diphtheriae toxin testing) is a send-away test.
Pertussis	A person with a cough of more than 2 weeks with at least one of: paroxysms of coughing, inspiratory whoop, or post-tussive vomiting without apparent cause	• NP swab (nasopharyngeal, viral transport medium) — for Bordetella pertussis PCR via the Molecular Laboratory (Section 11.7); collect as early in the illness as possible, ideally within the first 2–3 weeks of cough onset and before antibiotics are started • NP swab (Amies transport medium) — for culture, if PCR is unavailable or a second swab can be tolerated • PCR confirmation is the preferred method; culture sensitivity declines rapidly after the first week of illness and after antibiotic treatment • Validate DPT/pentavalent vaccination history and record on the surveillance form

23.3. ALERT THRESHOLDS FOR ACTION

Whether an event constitutes an outbreak depends on the number of cases normally seen in a population, considering factors such as season, changes in population size, and changes in the surveillance system (e.g. increase in reporting sites or recent training). For some diseases a single case requires urgent action; others require cases to exceed a threshold before investigation is triggered.

The alert thresholds below define when an event must be verified and, as necessary, investigated. Thresholds are regularly assessed and adjusted to improve sensitivity and specificity in detecting outbreaks.

Syndrome/disease	Alert threshold	Comments
Acute Fever and Rash (AFR)	1 case	Suspected measles or Zika
Prolonged Fever (PF)	1 case	
Influenza-Like Illness (ILI)	3 or more linked cases	
Watery Diarrhoea (WD)	3 or more linked cases	
Dengue/Zika/Chikungunya-Like Illness	1 case	Suspected dengue, Zika, or chikungunya
Acute Flaccid Paralysis (AFP)	1 case	Suspected polio or Guillain-Barre syndrome related to Zika

Syndrome/disease	Alert threshold	Comments
Neonatal Tetanus (NT)	1 case	
Diphtheria	1 case	
Pertussis	1 case	

In the context of disease elimination, the following diseases also require investigation based on a single suspected or confirmed (index) case:

- Malaria (in Tafea, Shefa, and Torba provinces)
- Tuberculosis
- Yaws
- Trachoma
- Leprosy
- Scabies
- Lymphatic filariasis (eliminated in Vanuatu)

24. SPECIMEN REFERRAL

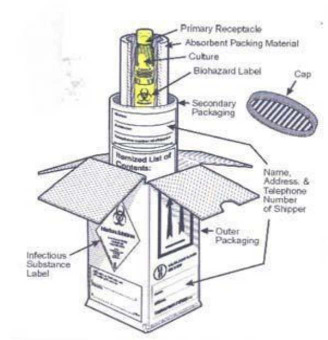
24.1. FROM OUTER ISLANDS (BY AIR)

Medical assistants are advised to consult the laboratory before collecting and packaging samples for referral.

24.2. PACKAGING OF SPECIMENS

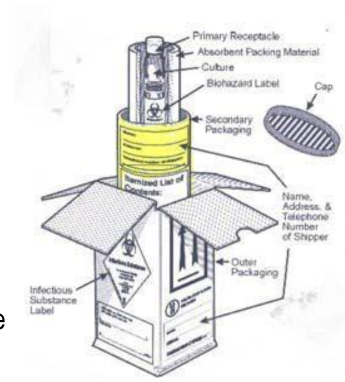
A triple-packaging system is required for all specimen shipment/transport to the Laboratory Department, consisting of three components:

PRIMARY RECEPTACLE (TUBE, VIAL, BOTTLE)



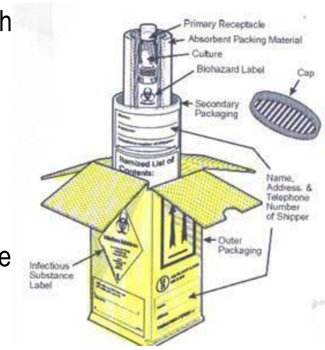
- The primary receptacle holds the specimen.
- The primary receptacle must be securely sealed and leak proof (screw top tubes must have a piece of waterproof tape around the top to prevent the top from coming loose in transit).
- The primary receptacle must be surrounded by absorbent material capable of taking up the entire liquid contents.
- The primary receptacle must be packed in the secondary receptacle in such a way that it will not break.

SECONDARY PACKAGING



- This is where the primary receptacle and the absorbent and cushioning material are placed.
- The secondary packaging must be leak proof and securely sealed.
- The secondary packaging must be placed in the outer packaging so that it does not move.
- The biohazard marking should be on the secondary receptacle and may be on the primary receptacle.
- Itemized list of contents:
  - An itemized list of contents is required. DO NOT place documents inside the secondary container.
  - The itemized list is placed OUTSIDE the secondary container. The laboratory test requisition form may serve this purpose. The form should also be placed OUTSIDE the secondary container.

OUTER PACKAGING



- The outer container is the receptacle into which the secondary packaging with cushioning materials are placed.
- The outer packaging must be rigid.
- The outer packaging bears the addressing information along with all required markings and labels. The full name and address of the shipper and the consignee must be on the outside packaging. The outside packaging must also have the name and telephone number of a person who is knowledgeable about the contents of the shipment. This is important emergency information in the event an exposure occurs during shipping.

For liquids, absorbent material in sufficient quantity to absorb the entire contents should be placed between the primary receptacle(s) and the secondary packaging so that, during transport, any release or leak of a liquid substance will not reach the outer packaging and will not compromise the integrity of the cushioning material. When multiple fragile primary receptacles are placed in a single secondary packaging, they should be either individually wrapped or separated to prevent contact between them.

24.3. SEND-AWAY (OVERSEAS) TESTING

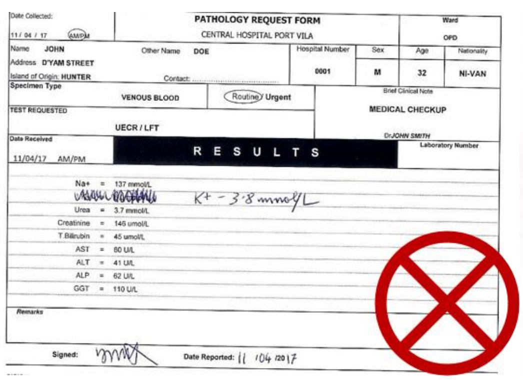
Tests marked SEND-AWAY in the department test tables above are not processed at VNH - they are referred out, and results take days to weeks rather than hours. Discuss timing with the relevant department before sending. If a test is needed that isn't listed anywhere in this handbook, contact the relevant department directly; they can advise whether it can be referred out.

Specimens from Operating Theatre, wards, or Reception that need further referral should be sent to the Laboratory Department Reception Unit, which coordinates onward shipment.

25. LABORATORY RESULTS AND REPORTS

All hard-copy results are available from the Laboratory Department Reception Unit's respective ward pigeonholes. Routine results are dispatched to ward pigeonholes twice daily (around 12:00 and before 16:00).

- Unexpectedly or markedly abnormal results are routinely telephoned to the concerned clinician.
- Hard copies of abnormal results are flagged to attract the attention of the Nurse-in-Charge and delivered to the ward promptly once verified by the duty scientist.
- Ward, OPD, and SOPD staff must sign the Result Register Log Book when collecting reports. Patients and family members are discouraged from collecting results directly, to protect confidentiality.
- All reports require two laboratory officers' signatures (performed-by and verified-by) before they may leave the laboratory; reports without verification signatures must not be removed.
- Corrections to a report must be made with a single straight line through the incorrect result, the corrected result written alongside, and signed with date and time — scribbled-over or overwritten results must not be accepted as evidence for patient management. See example below;



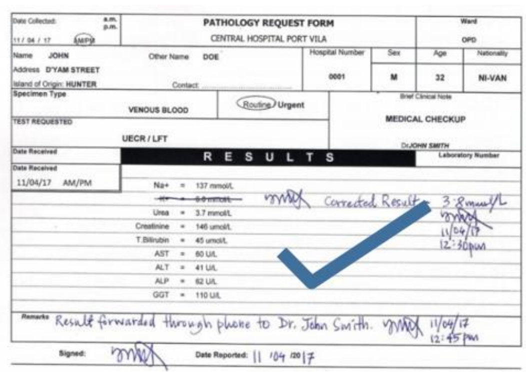
PATHOLOGY REQUEST FORM
CENTRAL HOSPITAL PORT VILA

Date Collected: 11 / 04 / 17 AM/PM
Ward: OPD
Name: JOHN Other Name: DOE
Address: D'YAM STREET
Island of Origin: HUNTER Contact: ....
Hospital Number: 0001 Sex: M Age: 32 Nationality: NI-VAN
Specimen Type: VENOUS BLOOD Routine / Urgent
TEST REQUESTED: UECR / LFT
Brief Clinical Note: MEDICAL CHECKUP Dr.JOHN SMITH
Date Received: 11/04/17 AM/PM
RESULTS
Laboratory Number

Na+ = 137 mmol/L
K+ - 3·8 mmol/L
Urea = 3.7 mmol/L
Creatinine = 146 umol/L
T.Bilirubin = 45 umol/L
AST = 60 U/L
ALT = 41 U/L
ALP = 62 U/L
GGT = 110 U/L

Remarks

Signed: Date Reported: 11 /04 /2017



PATHOLOGY REQUEST FORM
CENTRAL HOSPITAL PORT VILA

Date Collected: a.m. p.m. 11 / 04 / 17 AM/PM
Ward: OPD
Name: JOHN Other Name: DOE
Address: D'YAM STREET
Island of Origin: HUNTER Contact: ....
Hospital Number: 0001 Sex: M Age: 32 Nationality: NI-VAN
Specimen Type: VENOUS BLOOD Routine / Urgent
TEST REQUESTED: UECR / LFT
Brief Clinical Note: MEDICAL CHECKUP Dr.JOHN SMITH
Date Received
Date Received: 11/04/17 AM/PM
RESULTS
Laboratory Number

Na+ = 137 mmol/L
~~K+ = 6.0 mmol/L~~ Corrected Result 3·8 mmol/L 11/04/17 12:30pm
Urea = 3.7 mmol/L
Creatinine = 146 umol/L
T.Bilirubin = 45 umol/L
AST = 60 U/L
ALT = 41 U/L
ALP = 62 U/L
GGT = 110 U/L

Remarks: Result forwarded through phone to Dr. John Smith. 11/04/17 12:45pm

Signed: Date Reported: 11 /04 /2017

26. LABORATORY CONSUMABLES

VNH departments can order laboratory consumables (specimen collection materials etc) using the approved Laboratory Consumable Order Form (Annexure 6). Orders are placed by unit size, on **Thursdays only**, and must be authorized by a supervisor's signature.

ANNEXURE 1 — TRANSFUSION ADVERSE REACTIONS

Transfusion reactions can be classified in simple categories in order to help you to recognize them, understand their underlying causes and when they may occur, and know how to prevent, manage and record them. Acute complications of transfusion Acute transfusion reactions occur during or shortly after (within 24 hours) the transfusion. They can be broadly classified in the following three categories according to their severity and the appropriate clinical response.

ACUTE REACTIONS (DURING OR WITHIN 24 HOURS OF TRANSFUSION)

CATEGORY 1: MILD

- Mild hypersensitivity -- allergic, urticarial reactions.

CATEGORY 2: MODERATELY SEVERE

- Moderate-severe hypersensitivity (severe urticarial reactions).
- Febrile non-haemolytic reactions -- antibodies to white cells/platelets, antibodies to proteins (including IgA), possible early bacterial contamination, pyrogens.

CATEGORY 3: LIFE-THREATENING

- Acute intravascular haemolysis.
- Bacterial contamination and septic shock.
- Fluid overload.
- Anaphylactic reactions.
- Transfusion-associated lung injury.

DELAYED COMPLICATIONS

Transfusion-transmitted infections: HIV-1/2, HTLV-I/II, hepatitis B and C, syphilis, Chagas disease, malaria, cytomegalovirus, and other rare infections (e.g. human parvovirus B19, hepatitis A).

Other delayed complications (days, months, or years later): delayed haemolytic reaction, post-transfusion purpura, graft-versus-host disease, and iron overload in repeatedly transfused patients.

ANNEXURE 2 – LABORATORY REQUEST FORM

VCHL QUAL-7002.

L A B O R A T O R Y R E Q U E S T F O R M

Laboratory Department
Vila Central Hospital
PMB 9013
Port Vila, Vanuatu

G E N E R A L R E Q U E S T F O R M

WARD

PATIENT'S INFORMATION

First Name ______ Surname ______

Island/Country of Origin ______ National ID ______

Date of Birth ______ Address ______

Nationality ______ Contact ______

Hospital ID ______ Island of Residence ______

Sex ☐ Male ☐ Female Province of Residence ______

CLINICIAN'S INFORMATION

Full Name ______ Full Name ______

Date ______ Time ______ Contact ______

SPECIMEN INFORMATION

Sample Type

☐ Blood ☐ Routine

☐ Urine ☐ Urgent

☐ Other Sample:

CLINICAL INFORMATION

REQUESTING TEST

LABORATORY SECTION ONLY

VCHL ID:

Date received: Time received: ☐ AM ☐ PM

Received by: Section ID:

Date processed: Time processed: ☐ AM ☐ PM

Date

Processed by: Verfied by: Entered by:

Review all clinical and lab data carefully before requesting a TEST to avoid unnecessary procedures and delayed

Version: II

ANNEXURE 3 – CROSSMATCH REQUEST FORM

VCHL QUAL-7002.2

LABORATORY REQUEST FORM

Laboratory Department
Vila Central Hospital
PMB 9013
Port Vila, Vanuatu

TRANSFUSION REQUEST FORM

WARD

PATIENT'S INFORMATION

First Name ____ Surname ____

Island/Country of Origin ____ National ID ____

Date of Birth ____ Address ____

Nationality ____ Contact ____

Hospital ID ____ Island of Residence ____

Sex ☐ Male ☐ Female Province of Residence ____

CLINICIAN'S INFORMATION

Full Name ____ Signature ____

Date ____ Time ____ Contact ____

SPECIMEN INFORMATION

Sample Type

☐ Blood ☐ Routine ☐ **Urgent**

Date Collected Time Collected

REASON FOR TRANSFUSION

REQUESTING PRODUCT

☐ Blood Type ☐ FFP ☐ Cryo ☐ Packed Cells ☐ PRP ☐ Fresh Whole Blood

☐ Platelet Rich Concentration Date Required: ____

LABORATORY SECTION ONLY

VCHL ID:

Date received: ____ Time received: ____ ☐ AM ☐ PM

Received by: ____ Section ID: ____

Patient's Blood Type: ____ **Rh:** ____ Patient's Haemoglobin: ____ **g/dL**

DNR UNIT #	BLOOD TYPE	EXPIRY DATE	COMPATIBILITY	DATE DONE	COMMENT

Date processed: ____ Time processed: ____ ☐ AM ☐ PM

Remarks: ____

Processed by: ____ Verified by: ____ Authorized by: ____
Initial or Signature Initial or Signature Initial or Signature

Review all clinical and lab data carefully before requesting a crossmatch to avoid unnecessary procedures and delays!

Version: II

ANNEXURE 4 – TB REQUEST FORM

VCHL QUAL-7217

Laboratory Department
Vila Central Hospital
PMB 9013
Port Vila, Vanuatu

REQUEST *for* MTB/AFB

VCHL ID ______

PATIENT'S INFORMATION

First Name ______ Surname ______

Island/Country of Origin ______ National ID ______

Address ______

Date of Birth ______

Nationality ______ Contact ______

Hospital ID ______ Island of Residence ______

Sex ☐ Male ☐ Female Province of Residence ______

IMMEDIATE CONTACT'S INFORMATION

Full Name ______ Date of Birth ______

Island/Country of Origin ______ National ID ______

REFERRAL INFORMATION

Referred from: ☐ DOTS Center ☐ Hospital/Ward ______ *Print Name Clearly*

Referred by: ______ *Clinician Name (Print)* ______ *Clinician signature*

Date: ______ Specimen ID: ______ TB Reg ID: ______

SPECIMEN INFORMATION

Sample Type

☐ Sputum
☐ CSF
☐ Other Sample: ______

☐ First sample
☐ Second sample
☐ Third Sample
☐ Fourth Sample

REASON FOR EXAMINATION

☐ Diagnostic
☐ Suspect
☐ Follow up
☐ Other: ______

☐ Urgent
☐ Routine

LABORATORY SECTION ONLY

Date received: ______ Time received: ____ : ____ ☐ AM ☐ PM

Received by: ______ Department ID: ______

Quality and Appearance of specimen:

☐ Good quality ☐ Watery ☐ Saliva ☐ Inadequate ☐ Unsatisfactory ☐ Blood stained

☐ Muco-purulent ☐ Other: ______

Date processed: ______ Time processed: ____ : ____ ☐ AM ☐ PM

Remarks: ______

Processed by: ______ *Initial or Signature* Verified by: ______ *Initial or Signature* Entered by: ______ *Initial or Signature*

For more information, please contact +678 7337919 or +678 22100 VOIP: 1987

REQUEST FOR MTB/AFB

Version: II

ANNEXURE 5 – LABORATORY SURVEILLANCE FORM

VCHL QUAL-7219

Laboratory Notifiable Disease REQUISITION

LABORATORY DEPARTMENT
Vila Central Hospital
PMB 9013
Port Vila

REQUEST *for*
Acute Fever & Rash (AFR) - Influenza-Like Illness (ILI) - Prolonged Fever - Diarrhoea - nCovid-19

Requisition ID:

SENDER'S INFORMATION

Case Investigation Form ID

Sender's name & address
Phone Contact
Office Contact
Position
Signature
Email

PATIENT'S INFORMATION

Surname/Code
Island/Country of Origin
First name/Code
Current Address
Hosp Number
National ID
Permanent Address
Gender ☐ male ☐ female
Date of Birth
Age
Contact
Nationality
Passport Number
Email

TRAVEL HISTORY

Foreign Travel? ☐ Yes ☐ No
Name of **Country**
Purpose of travel
Place visited
Travel Date
☐ Urban area ☐ Open country ☐ Forests
☐ Rural area ☐ *Water contact** ☐ *Animal contact**
Date returned
**Specify*
Relevant Occupational History

CLINICAL INFORMATION

Onset Date of Illness

Pregnant? ☐ Yes ☐ No

Muscular Skeletal System
☐ Aches & Pain
☐ Fatigue/Tiredness
☐ Headache
☐ Joint pain
☐ Muscle pain
☐ Mayalgia
☐ Malaise

GIT System
☐ Diarrhoea
☐ Hepatic failure
☐ Jaundice
☐ Nausea
☐ Vomiting
☐ Abnormal LF[1]

Respiratory System
☐ Koplik's spot
☐ Cough-dry
☐ Cough-productive
☐ Flu-Like Illness
☐ Pneumonia
☐ Runny nose
☐ Shortness of breath
☐ Sore throat

Nervous System
☐ Encephalitis
☐ Fever ___ °C
☐ Inflamation
☐ Meningtis
☐ Typhoid fever

Vascular System
☐ Spontaneous bleeding
☐ Thrombocytopenia

Oculi System
☐ Conjunctivitis
☐ Eye pain

Others
☐ Hospitalization
☐ Asymtomatic
☐ Occipital, Cervical & Aurical Lymph nodes
☐ Abnormal RF[2]
☐ Rash
☐ **Other conditions/symptoms/Rash**

1 Liver function test, 2 Renal function test

Surveillance Purpose
☐ Contact of a case
☐ Suspected while seeking care at health facility
☐ Suspected at point of entry
☐ Suspected during quarantine
☐ Severe acute respiratory illness (SARI)
☐ Unknown

SAMPLE INFORMATION

☐ Serum/Clotted Samples
☐ EDTA Whole Blood
☐ CSF
☐ Urine *(Only send with paired sera)*
☐ Other *(please specify)*
☐ Stool
☐ Blood culture
☐ Dried Blood Spot
☐ Nasopharyngeal Swab
☐ Oropharyngeal Swab
☐ BAL Fluid
☐ Tracheal Aspirate
☐ Post Mortem

Collection site/ward
Collected by
Collection Date
Collection Time

All samples must be sent to the Laboratory in accordance to the Class 6, Division 6.2 Infectious Substance of the UN transportation of Dangerous Goods Regulation

TEST REQUESTED

☐ Dengue ☐ Leptospirosis ☐ Measles ☐ MCS ☐ Covid-19
☐ Malaria ☐ Rubella ☐ Rota virus ☐ Pertusis ☐ Other tests *(Please specify)*
☐ Influenza ☐ Chikugunya ☐ Adeno virus ☐ Diptheria

LABORATORY SECTION ONLY

Date received: ____________ Time received: ____ : ____ ☐ AM ☐ PM Received by: ____________

Department ID: ____________ Date processed: ____________ Time processed: ____ : ____ ☐ AM ☐ PM

Processed by: ____________ *Initial or Signature* Verified by: ____________ *Initial or Signature*

Remarks:

Version: III
Updated: August 2025

For more information, please contact +678 7309510, +678 7736759 or +678 22100 Ext: 1949

ANNEXURE 6 – LABORATORY CONSUMABLE ORDER FORM

LABORATORY DEPARTMENT
Vila Central Hospital
PMB 9013
Port vila
PHONE: +678 22100 Ext: 1949, 1948

DOMESTIC ORDER FORM

Laboratory Department

VCHLQUAL-7002.9

Requesting Department/Ward: ____________ **Address:** VILA CENTRAL HOSPITAL

ID	ITEM DESCRIPTION		UNIT SIZE	QUANTITY ORDERED AND DELIVERED					
				DATE		DATE		DATE	
	Item Actual Name	Common Name		ORDERED dd/mm/yy	DELIVERED dd/mm/yy	ORDERED dd/mm/yy	DELIVERED dd/mm/yy	ORDERED dd/mm/yy	DELIVERED dd/mm/yy
1	BIOHAZARD BAG	SPECIMEN TRANSPORT BAG	Bag						
2	BROWN TOP CONTAINER	60ml 'STOOL' CONTAINER	Container						
3	BLUE TOP TUBE	Na CITRATE (INR Tube)	1 Rack = 50 Tubes						
4	BLACK TOP STERILE CONTAINER	CSF TUBE	Container						
5	CULTURE SWAB TRANSPORT SYSTEM	SWAB TRANSPORT SYSTEM	1 pkt = 50 swabs						
6	FROSTED END M SLIDES	MICROSCOPE SLIDES	1 pkt = 50 slides						
7	MICRO TUBE EDTA K3	PURPLE CAP PAEDS	1 Rack = 100 Tubes						
8	MICRO TUBE SERUM	RED CAP	1 Rack = 100 Tubes						
9	EDTA TUBE - PURPLE TOP	4ml PURPLE TOP TUBE	1 Rack = 50 Tubes						
10	RED TOP TUBE PLAIN	BD VACUTAINER-4ml RED TOP TUBE	1 Rack = 100 Tubes						
11	YELLOW TOP CONTAINER	60ml 'URINE' CONTAINER	Container						
12	AEROBIC BC BOTTLE	BLOOD CULTURE SYSTEM	Bottle						
13	ANAEROBIC BC BOTTLE	BLOOD CULTURE SYSTEM	Bottle						
14	PAEDIATRIC BC BOTTLE	PAEDIATRIC BLOOD CULTURE SYSTEM	Bottle						
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									
26									

Orders will NOT be prepared unless all orders are authorized by a certified & registered supervisor.

Complete the order form accordingly.
Where:
1. Requesting Department = Wards (Medical, Surgical,...)
2. Address = Vila Central Hospital
3. Date Ordered = Date when order is being done (dd/mm/yy)
4. Date Delivered = Date when Laboratory Dept. prepared order (dd/mm/yy)
5. Requesting Officer Initial = Initial of Officer placing the Order
6. Requesting Officer Position = Position of the officer placing the Orders
7. Authorizing Officer Initial = Initial of the Officer In-Charge in the requesting Department
8. Authorizing Officer Position = Usually the requesting Department Officer In-Charge

Order using UNIT SIZE ONLY

For Requesting Department ONLY

	REQUESTING OFFICER	REQUESTING OFFICER	REQUESTING OFFICER
Initial:			
Position:			
	AUTHORIZING OFFICER	AUTHORIZING OFFICER	AUTHORIZING OFFICER
Initial:			
Position:			

	For Laboratory Department Use Only		
Order Prepared By:			
Authorized By:			

THIS FORM IS INTENDED TO BE USED BY WARDS/DEPARTMENTS AT VILA CENTRAL HOSPITAL ONLY FOR ORDERING CONSUMABLES AT THE LABORATORY DEPARTMENT

Tittle	*VCH LABORATORY DEPARTMENT DOMESTIC ORDER FORM*	Date Issued: **March 2019** Next Review Date: **March 2020**
Authored by: *SERO Kalkie*	Verified By: *Ms. GAVIGA Justina*	Authorized By: *Mr. Junior George PAKOA*

ANNEXURE 7 – LABORATORY COMPLAINT & COMPLIMENT FORM

VNH LABORATORY QUALITYMANUAL	COMPLAINTS	**QUAL. 1037** Page **4** of **4**

Vanuatu National Hospital
Laboratory Department

VANUATU NATIONAL HOSPITAL
LABORATORY DEPARTMENT
PMB 9013, Port Vila, Vanuatu
Telephone: 22100

LABORATORY COMPLIMENT & COMPLAINT FORM

Time:	Date: / /	How was the incident reported: ☐ Written ☐ Verbal
Person Reporting the incident: ______________________		Contact No.____________________ Email Address:__________________
Who they are representing:		
Name of staff Member receiving the information: ______________________		
Patient Name:		
Patient date of Birth:		Lab ID #(s):
Specimen collected date: / / Time collected: __________		Location: Collected by:

Summary and Nature of incident(Include date, time, & place)

Action Taken(Date/time)

In Progress By:	Date:	Action by:	Date
Referred to:	Date:	Filed for reference by:	Date

ANNEXURE 8 – PATHOLOGY REQUEST FORM